

A Parameterized Training Corpus for the Dirichlet–Neumann Operator

Design, implementation checks, and direct comparison with `combined_dataset_v9`

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Abstract

The historical training archive for the periodic Dirichlet–Neumann operator combines generators with different sampling, filtering, target, and snapshot rules. It is therefore difficult to separate the effects of physical coverage, numerical labels, and source-dependent row counts.

This document specifies the completed replacement corpus with four physical families: fifth-order Stokes states, periodic Tanaka profiles, Benjamin–Feir sideband states, and resolved-band JONSWAP/TMA random seas. All four use one delivered order-six DNO target on the Fourier band $|k| \leq 128$, case-level splitting, equal case weight, and reproducible case records. The three trajectory families use the same residual-controlled two-stage, fourth-order Gauss–Legendre method with step 0.01. Tanaka and Benjamin–Feir use 1024 internal points and the band $|k| \leq 256$; JONSWAP/TMA uses 2048 internal points and $|k| \leq 704$. All states and labels are restricted to the common delivered band. Tanaka uses DNO order six and a power-36 Hou–Li step filter. Benjamin–Feir revision 4 and JONSWAP/TMA revision 4 use DNO order four and a sharp internal cutoff; JONSWAP/TMA revision 4 restricts its initial spectrum to $0.5\omega_p \leq \omega \leq 2.5\omega_p$ and first undergoes the nonlinear adjustment defined below. A trajectory case is retained only when its declared calculation returns a complete admissible trajectory. Static Stokes cases instead require a finite state in the declared Stokes support. Airy, manufactured-DNO, and sparse-Fourier cases are reserved for validation.

The four samplers, constructors, common target, one-arm trajectory executor, bounded accepted-case scheduler, restart records, combined dataset views, and case-balanced loader are implemented and source-authenticated. The final release contains 73728 accepted cases from 74045 attempts, 317 zero-row rejections, and 7686144 retained rows. Every family contains 16384 training, 1024 validation, and 1024 test cases. The superseded Tanaka population and all pre-bulk gates remain diagnostic evidence outside the release. No reported model has been trained on the completed corpus.

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Reading guide. The document proceeds from the DNO target to the four initial-condition formulas, reference rollouts, whole-case acceptance, and balanced training batches. Sections [1](#), [2](#), [4](#), [6](#), and [8](#) define that chain. The difference from the historical archive, the authenticated release evidence, and the remaining model-training work are stated separately. The detailed **v9** audit is in [Appendix A](#).

Implementation status

The replacement corpus and its cumulative views are complete. The following table distinguishes final family releases from historical diagnostic gates and the still-future model-training work:

Item	Status	Evidence
Discrete target	Frozen	$L = 2\pi$, $g = 1$, order six, $N = 1024$, pad factor eight, and common delivered band $ k \leq 128$; fixed-band, mean, translation, and dtype tests pass.
Finite-depth Stokes	Revision-2 production complete	The four-category sampler enforces the conservative condition $Ur_+ \leq 26$. At $(N, M, p, K) = (1024, 6, 8, 128)$, one revision-2 case per category passed the complete sampled-specification, construction, target, quality, archive, and dataset-view transaction through the bounded quota executor. Production contains 16384 training cases and 1024 cases in each of validation and test; these completed artifacts are preserved.
Tanaka	Corrected revision-3 production complete	The eleven-category direction-aware periodic-gap sampler and tangent-Hermite constructor are connected to the common transaction. Revision 3 draws amplitudes first, samples depth conditionally so that the delivered band resolves the narrowest crest with ratio at least ten, evolves an internal $ k \leq 256$ state with a power-36 Hou–Li filter, and delivers only P_{128} fields and labels. Only a finite negative potential radicand is outside the declared construction; nonfinite algebra or an invalid wave speed stops generation. The two gate chunks accepted all eleven requested cases without rejection; their 2200 validation-tagged rows are excluded from the corpus. A historical run contains 4096 trajectories, but its old solitary-wave solve inflated one small component in 64 cases. The corrected solve widens the amplitude bracket, uses 48 bisections, and checks every realized crest height. The 4096 old parameter specifications are not reused. The fresh release contains 16384 training cases and 1024 cases in each held-out split, retaining all 18432 attempts and storing 3686400 rows.
Benjamin–Feir	Revision-4 production complete	The 66 integer pairs are parameter categories. The support bounds the leading-NLS focused-envelope steepness and makes the amplitude of each sideband five to ten percent of the carrier first-harmonic amplitude. Support-validation stream 934 accepted all 66 selected cases. The direct production-law stream 935 then accepted all 66 first proposals under the sharp order-four $K = 256$ evolution and common order-six P_{128} label. Production contains 16384 training cases and 1024 cases in each of validation and test, retained from 18455 attempts.
Random sea	Revision-4 production complete	The 27 categories store both phase arrays. Revision 4 restricts the constructed spectrum to $0.5\omega_p \leq \omega \leq 2.5\omega_p$ and requires that interval to fit in the delivered band. The dedicated path adjusts for 20 peak periods on 2048 points, hands off the full order-four $K = 704$ state, and keeps the common 1024-point order-six P_{128} label. GL2 uses at most five fixed-point updates per stage. The $K = 704$ versus $K = 768$ sentinel passed the global whole-field test, and the predeclared revision-4 gate retained 540 cases from 548 attempts. The bulk release retained 18432 cases from 18726 attempts, recorded 294 zero-row rejections, and stored 294912 rows.
Common trajectory transaction	Implemented	One declared float64 numerical path with GL2 step 0.01, saved spacing 0.08, the declared family–revision spatial contract, stage residual telemetry, dense saved-frame internal health telemetry where required, post-acceptance time selection, a complete-case writer, and a split-aware loader are executable and tested. The 0.01/0.005 pair remains a validation calculation and is not run for each production case.
Accepted-quota recovery	Bounded run specification implemented	The driver reconstructs checked accepted counts, replays proposal-only and proposal-plus-shard transaction states, and admits one writer. A category requesting Q_i accepted cases has at most $4Q_i$ durable attempts. Pending work at the ceiling is replayed; a resolved deficient category at the ceiling terminates the family run instead of silently changing its population.
Combined view and loader	Complete and authenticated	The view verifies transaction hashes without copying field arrays. It requires one revision per family and checks numerical and source identity separately for each (family, revision) pair. In each training epoch the loader selects one stored time from every accepted case;

1 Dataset organization

Let $L > 0$ be the horizontal period and $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$ the periodic interval, with horizontal coordinate $x \in \mathbb{T}_L$. Each training case stores a real surface elevation $\eta_0(x)$, a real surface-potential trace $\xi_0(x)$, and a constant mean depth $h > 0$. Write $x_\ell = \ell L/N$ for the numerical nodes of the relevant family. Each case also stores whether it was generated as a Stokes, Tanaka, Benjamin–Feir, or random-sea case and every numerical parameter required to reconstruct that initial state. The four constructions are defined explicitly in Section 4. Every accepted initial state satisfies

$$\int_{\mathbb{T}_L} \xi_0(x) \, dx = 0, \quad \min_{\ell} \{h + \eta_0(x_\ell)\} > 0. \quad (1)$$

The first condition fixes the irrelevant additive constant in ξ_0 ; the second excludes an initial dry point. Depth is stored with the case and held fixed during a rollout.

The dataset rules are:

1. Generate the same accepted case count from each family. Balance the coarse parameter ranges listed in Section 4.
2. First require the sampled specification to lie in its declared family support and its declared constructor to return a well-defined periodic initial state. For Tanaka revision 3, this support includes the resolution condition $R \geq 10$ defined in Section 4. A finite negative Tanaka radicand is outside the declared construction; a nonfinite radicand or an invalid wave speed stops generation. The trajectory adapters also require the assembled initial arrays to be finite, the depth to be positive, and $h + \eta_0 > 0$ on the grid. Failure of these constructor invariants normally stops generation rather than becoming an ordinary zero-row draw. The declared JONSWAP/TMA phase-realization check is the exception: nonfinite assembled fields or a nonpositive realized water column produce a recorded zero-row construction rejection for that case, which is replaced in the same category while valid batch companions continue. Other unclassified constructor failures stop generation. For a trajectory family, retain the complete case only if its declared fixed-step numerical path produces a complete admissible trajectory as defined in Section 6; static Stokes cases retain one well-defined state in their declared support. A failed trajectory contributes zero frames.
3. At each training epoch, choose one stored time uniformly from every accepted training case and then batch the selected rows. Because the accepted case count is the same in all four families, every family and every case has equal epoch weight. Validation uses one fixed, deterministic stored time from every validation case.
4. Keep Airy, manufactured-DNO, and sparse-Fourier cases out of training. They are diagnostic tests only.

There is no production rejection rule based on sign changes, slope, extrema, absolute target size, or visual appearance. Hamiltonian drift is a declared numerical condition only for autonomous Benjamin–Feir revision-4 and JONSWAP/TMA revision-4 trajectories, as defined in Section 6. The replacement corpus uses low-steepness Stokes states instead of a separate single-mode family, and a standard finite-depth random-sea spectrum instead of the old Gaussian sea. Its shallow mixed-mode coverage is tested as specified in Section 9. The existing v9 archive remains the experimental baseline.

2 Learning problem and discrete target

Let $g > 0$ denote gravitational acceleration; the generated corpus uses the nondimensional value $g = 1$. For each integer Fourier index n , define the wavenumber $k_n = 2\pi n/L$. For any integrable L -periodic function u , use the Fourier convention

$$\widehat{u}_n = \frac{1}{L} \int_{\mathbb{T}_L} u(x) e^{-ik_n x} dx, \quad u(x) = \sum_{n \in \mathbb{Z}} \widehat{u}_n e^{ik_n x}. \quad (2)$$

For an N -point grid, define the spacing $\Delta x = L/N$ and nodes $x_j = j\Delta x$ for $j = 0, \dots, N-1$. The normalized continuous and grid norms are

$$\|u\|_{2,L}^2 = \frac{1}{L} \int_{\mathbb{T}_L} |u|^2 dx, \quad \|u\|_{2,N}^2 = \frac{1}{N} \sum_{j=0}^{N-1} |u(x_j)|^2. \quad (3)$$

Unless a subscript is shown, $\|\cdot\|_2$ means the corresponding normalized norm on the representation being discussed. Fix a maximum retained wavenumber $K > 0$. Let P_K be the fixed-band projection, let $\mathcal{J}_K$ be the set of its retained positive Fourier indices, and let the indicator $\mathbf{1}_{\{\cdot\}}$ equal one when the condition in braces holds and zero otherwise:

$$\widehat{P_K u}_n = \mathbf{1}_{\{|k_n| \leq K\}} \widehat{u}_n, \quad \mathcal{J}_K = \{n \in \mathbb{Z} : n \geq 1, k_n \leq K\}. \quad (4)$$

The projection acts componentwise when its argument is an ordered pair of fields. Define the mean-removal operator $P_0^\perp$ and the mean-zero band projection P_K° by

$$P_0^\perp u = u - \widehat{u}_0, \quad P_K^\circ = P_0^\perp P_K. \quad (5)$$

Thus P_K° both restricts the band and fixes the zero-mean gauge. It is used for DNO outputs and for the ξ component of the vector field; P_K alone is used for η , whose mean is the conserved mass. The frozen target cutoff is $K = 128$. The revision-4 random-sea constructor also restricts its input spectrum, but does so by the case-relative frequency indicator in (58), not by a fixed wavenumber taper. That restriction changes the constructed initial condition and must not be conflated with P_K .

For a surface elevation $\eta(x)$ and depth $h > 0$, let y denote the vertical coordinate and define the fluid domain

$$\Omega_{\eta,h} = \{(x, y) : x \in \mathbb{T}_L, -h < y < \eta(x)\}. \quad (6)$$

For surface Dirichlet data ξ , let φ solve

$$\Delta \varphi = 0 \quad \text{in } \Omega_{\eta,h}, \quad \varphi_y(x, -h) = 0, \quad \varphi(x, \eta(x)) = \xi(x). \quad (7)$$

The physical Dirichlet–Neumann operator is

$$G(\eta; h)\xi = (\varphi_y - \eta_x \varphi_x)_{y=\eta(x)}. \quad (8)$$

Let $D = -i\partial_x$, so $|D|$ is the Fourier multiplier with symbol $|k_n|$. For each integer $m \geq 0$, let G_m be the Craig–Sulem term homogeneous of degree m in η , and let M be the largest retained order. The current solver uses

$$G_M(\eta; h)\xi = \sum_{m=0}^M G_m(\eta; h)\xi, \quad G_0(h) = |D| \tanh(h|D|). \quad (9)$$

In particular,

$$G_1(\eta; h)\xi = -G_0(h)(\eta G_0(h)\xi) - \partial_x(\eta \partial_x \xi). \quad (10)$$

This formula fixes the meaning of the analytic $G_0 + G_1$ backbone referenced later in the diagnostic discussion.

We distinguish the physical operator in (8) from its fully discrete evaluation. Let $\mathcal{G}_M^{N,p}$ denote the order- M pseudospectral recursion on N points with padding factor p . The frozen common learning target is

$$\boxed{G_{\text{ref}}(\eta; h)\xi = P_K^\circ \left[\mathcal{G}_M^{N,p}(P_K \eta; h)(P_K \xi) \right]}. \quad (11)$$

The frozen paper target uses

$$L = 2\pi, \quad g = 1, \quad M = 6, \quad N = 1024, \quad p = 8, \quad K = 128. \quad (12)$$

The evaluation in (11) is performed in float64 and only the final arrays are stored in float32. Statements that a constructed field has zero mean refer to the float64 value before storage; float32 conversion preserves the condition only to rounding error.

JONSWAP/TMA is evolved on a 2048-point grid, so its stored state requires one explicit grid transfer. For a field sampled on N' points define

$$\hat{f}_k^{(N')} = \frac{1}{N'} \sum_{j=0}^{N'-1} f\left(\frac{2\pi j}{N'}\right) e^{-2\pi i j k / N'}. \quad (13)$$

For every retained JONSWAP/TMA state, the coefficients of η and ξ with $|k| \leq 128$ are copied to a 1024-point grid and reconstructed there. The stored target is then recomputed with (11). The internal order-four value of $G(\eta; h)\xi$ is never resampled and is not a learning label.

Equation (11) says exactly which discrete formula the model is trained to reproduce. A claim that this formula approximates the physical G requires separate convergence evidence in M , N , and p . Fixed- M agreement between several grids does not measure Craig–Sulem truncation error. The implementation also tests that unresolved input modes do not change any delivered field, that $G_{\text{ref}}(\eta; h)\xi$ has zero mean, and that the target commutes with translations on the periodic grid. The manuscript therefore claims approximation of this declared discrete order-six target, not a measured continuum-DNO error.

For an accepted case, let t denote time and write the reference state as $z(t) = (\eta(t), \xi(t), h)$. Each stored training example is

$$(P_K \eta(t), P_K \xi(t), h, G_{\text{ref}}(\eta(t); h)\xi(t)). \quad (14)$$

The complete case, rather than the individual stored example, is the unit used for rejection and train/validation/test splitting.

3 Why regenerate the corpus

The current v9 archive is a useful performance baseline, but its sources do not share one data rule. They contribute different numbers of snapshots per physical case, use different output filtering and rejection rules, and do not always store enough information to reconstruct a rejected attempt. Consequently, a change in performance cannot be attributed cleanly to the initial-condition family alone.

The replacement corpus is defined in terms of complete case records. Every family uses the same delivered target, case-level split, batch weighting, GL2 method, and complete-trajectory requirement. Its internal spatial execution contract is fixed separately for each family and revision: Tanaka revision 3 uses the filtered order-six buffer defined below, whereas Benjamin–Feir revision 4 uses an order-four sharp buffer through $|k| = 256$ on 1024 points and JONSWAP/TMA uses one through $|k| = 704$ on 2048 points. Static Stokes cases instead use their declared parameter support, the frozen target, and generator-level spatial validation; they are not assigned a fictitious time-integration gate. Appendix A gives the exact v9 row counts, historical parameter ranges, and formulas needed for comparison; none of those old source names is treated as a new physical family.

4 Parameterized initial conditions

This section gives an explicit initial-state formula and parameter list for each family. The complete parameter list is written before integration, for both accepted and rejected attempts. Numerical formula versions, L , g , N , spectral restriction, and solver tolerances belong to a common construction configuration rather than being repeated in every case record. Here $\text{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. A positive variable is log-uniform on $[a, b]$ when its logarithm is uniform on $[\log a, \log b]$. The notation $\text{Dirichlet}(1, \dots, 1)$ denotes the uniform distribution on the simplex $\{(w_1, \dots, w_m) : w_j \geq 0, \sum_j w_j = 1\}$. Unless an endpoint convention is written explicitly, closed brackets describe the closure of the continuous support. The ordinary floating-point uniform draw includes its lower endpoint and excludes its upper endpoint. Explicit endpoint conventions in an individual sampling formula—including half-open phase and center intervals and the mixed open–closed Benjamin–Feir steepness draw—override that default. The distinction has probability zero for the continuous law but is recorded for exact replay.

4.1 Fifth-order Stokes states

Choose an integer carrier mode $n \geq 1$, amplitude parameter $a > 0$, phase $\phi \in [0, 2\pi)$, depth $h > 0$, and branch $r \in \{\text{finite}, \text{deep}\}$. Set $k = k_n$ and define the steepness $\epsilon = ka$. Let $\mathcal{S}_5$ denote the deterministic fifth-order Stokes constructor used by the project. It evaluates the frozen finite-depth coefficient table when $r = \text{finite}$ and its deep-water limit when $r = \text{deep}$; the fifth-order expansion and its range of application are discussed by Fenton and by Zhao and Liu [2, 1]. Define

$$(\eta_0, \xi_0) = \mathcal{S}_5(h, n, a, \phi, r). \quad (15)$$

The implemented generator contains the full coefficient table and its version identifier. Each Stokes case records (h, n, a, ϕ, r) and that formula version. The mean of ξ_0 is fixed to zero because adding an x -constant does not change the DNO. Finite and deep are parameter strata, not different physical families. The JAX constructor has also been checked against 21 snapshots produced independently by the collaborator’s MATLAB generator: relative discrepancies are at most 1.1×10^{-14} in η , 1.0×10^{-14} in ξ , and 9.0×10^{-13} in the order-six DNO output. Thus the shallow-support restriction below addresses the range of the perturbation formula, not a MATLAB-to-JAX transcription error. The sampling population has four parameter categories: the Cartesian product of the finite and deep branches with the two steepness intervals

$$0.005 \leq ka < 0.03 \quad \text{and} \quad 0.03 \leq ka \leq 0.15, \quad (16)$$

with the common amplitude interval $a \in [0.000766, 0.011494]$. On the finite branch,

$$n \in \{14, \dots, 26\}, \quad 0.02 \leq h \leq 1.5, \quad 0.5 \leq kh \leq 5.$$

On the deep branch,

$$n \in \{1, \dots, 20\}, \quad 4 \leq h \leq 50, \quad kh \geq 5.$$

Within a preassigned branch–steepness category, first list the integer modes for which the depth, amplitude, and steepness intervals have nonempty intersection. Choose uniformly from that finite list. Conditional on the mode, choose h log-uniformly on its allowed depth interval, choose ϕ uniformly on $[0, 2\pi)$, and choose a uniformly on the intersection of the common amplitude interval with the assigned steepness interval. Only feasible modes enter the list, and an attempted case never changes its preassigned branch or steepness category.

This population omits both the small fraction of archived deep-Stokes states with $ka < 0.005$ and the $n = 1, 4 \leq h < 5$ corner rather than claiming that it preserves the entire historical support. The low- ka category is part of the same Stokes family and formula limit, not a fifth training family. The fifth-order state approaches its Airy limit as $a \rightarrow 0$. Moreover, the paper model evaluates $G_0 + G_1$ analytically and constrains its learned remainder to begin at $O(\eta^2)$; duplicating that limit with hundreds of thousands of standalone linear states is therefore not the preferred use of the label budget. The Airy limit is tested independently in Section B.

The finite-depth coefficient table is a perturbation expansion and is not uniformly accurate over every pair (kh, ka) in the rectangular historical sampling range. Let

$$H = \max_{x \in \mathbb{T}_L} \eta_0(x) - \min_{x \in \mathbb{T}_L} \eta_0(x) \quad (17)$$

be the physical crest-to-trough height of the constructed profile, and let $\lambda = 2\pi/k = L/n$ be its carrier wavelength. Define the Ursell number

$$\text{Ur} = \frac{H\lambda^2}{h^3}. \quad (18)$$

The implemented series is $\eta_0(x) = \sum_{m=1}^5 E_m \cos(m\theta)$, where $\theta = kx + \phi$ and E_m is the coefficient of the m th elevation harmonic. Let H_{4096} be the largest sampled elevation minus the smallest sampled elevation on 4096 equally spaced phases, and define

$$B_2 = \sum_{m=1}^5 m^2 |E_m|, \quad H_+ = H_{4096} + \frac{B_2}{4} \left(\frac{2\pi}{4096} \right)^2. \quad (19)$$

Since $|\text{d}^2\eta_0/\text{d}\theta^2| \leq B_2$, Taylor’s theorem in the phase coordinate at the true maximum and minimum gives $H \leq H_+$. Define the conservative value $\text{Ur}_+ = H_+\lambda^2/h^3$. For the finite-depth branch, the sampling domain requires

$$\boxed{\text{Ur}_+ \leq 26}. \quad (20)$$

Fenton identifies $\epsilon/(kh)^3$ as the effective shallow-water expansion parameter for Stokes theory [2]. Zhao, Wang, and Liu use $\text{Ur} = 26$ to delimit the fifth-order Stokes regime from the shallower cnoidal regime [3]. Since the constructed wave height satisfies $H = 2a[1 + O(\epsilon^2)]$,

$$\text{Ur} = 8\pi^2 \frac{ka}{(kh)^3} [1 + O((ka)^2)]. \quad (21)$$

Thus (20) reduces at leading order to $ka \lesssim 0.32929(kh)^3$. The implemented decision uses the upper bound (19), not this leading approximation. It is evaluated from the analytic initial profile before integration and is independent of the sampled phase. The phase ϕ is sampled directly and held fixed; it is not inferred from a time whose nonlinear frequency would change when a changes. The

constructor removes the spatial mean of ξ_0 , which fixes the irrelevant constant-potential gauge. A rejected amplitude is resampled within its preassigned category; it is not clipped to the boundary. After the initial amplitude draw, at most 1000 additional amplitude redraws are allowed. Every rejected amplitude and its corresponding Ur_+ value are stored with the accepted case after a successful redraw. If the allowed redraws are exhausted, the ledger instead records the fixed mode, depth, phase, and category, together with every attempted amplitude and Ur_+ value. The executor commits that record as a zero-row out-of-support attempt and schedules a replacement in the same category. The family run stops only if the outer attempt ceiling in (69) is exhausted before the accepted-case quota is filled.

An independently seeded historical 256-case finite-depth diagnostic panel contained four nonfinite reference trajectories and three finite trajectories whose maximum recorded relative Hamiltonian drift exceeded 10^{-3} . All seven satisfy $80.50 \leq Ur_+ \leq 161.82$. The support condition also excludes 18 of the other 249 trajectories. This comparison diagnoses the sampled Stokes support; it does not turn the Ursell bound into a classifier of numerical failure: a general initial state can evolve finitely even when its intended fifth-order traveling-wave interpretation is outside the recommended regime. Historical case 98 has conservative $Ur_+ = 96.88$ and leading approximation 61.87, so both identify the sharp profile as an unsupported parameter draw. The previously considered ratio of correction-harmonic amplitudes to the fundamental remains a descriptive diagnostic only; no cited Stokes-wave analysis recommends it as a support boundary.

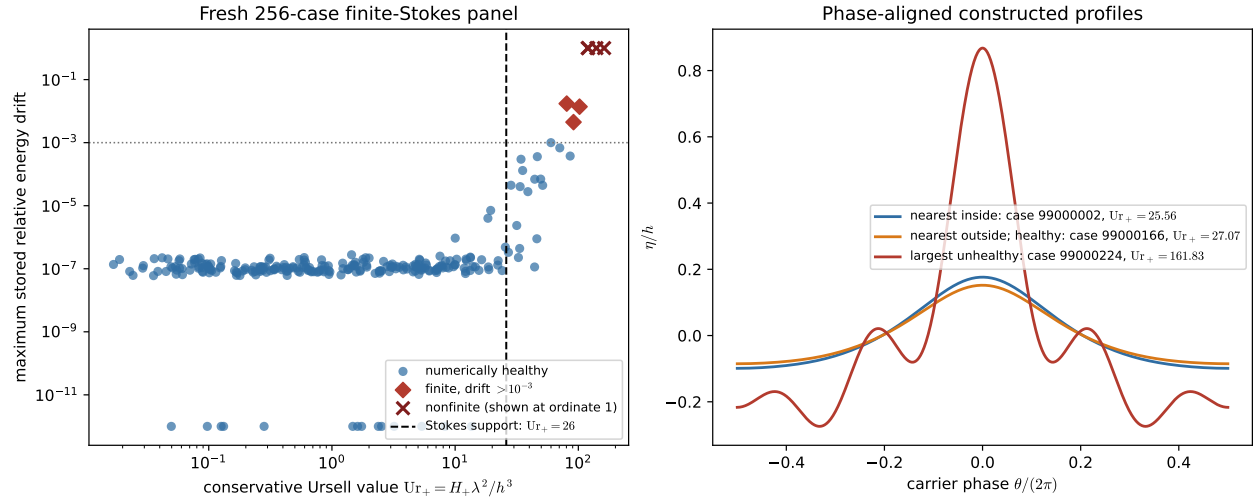


Figure 1: Historical finite-depth fifth-order Stokes diagnosis. The left panel places independently generated cases against the support boundary $Ur_+ = 26$ and distinguishes nonfinite trajectories and finite trajectories whose recorded relative Hamiltonian drift exceeds 10^{-3} . The right panels show the nearest cases on either side of the boundary and the largest flagged profile. The vertical line defines the adopted family support; it is not fitted as an exact classifier of integration failure.

Any enlargement beyond the current Stokes amplitude and depth support is a separate experiment. In particular, the old manufactured single-mode support ($n = 1, \dots, 20$ and kh as low as 0.02) is much broader than the finite-depth Stokes support and is not silently relabeled as Stokes data.

This is a fifth-order analytic approximation. It must not be described as the numerically exact Fenton Stokes construction used in the long-time tests of Xu and Guyenne [4].

4.2 Periodic Tanaka-profile sums

Let $\alpha = a/h$ be the ratio of crest amplitude a to depth h , and let X denote the dimensionless horizontal coordinate. Following Tanaka's solitary-wave formulation [5], the frozen unit-depth solve at amplitude ratio α returns a dimensionless elevation $\bar{\eta}_\alpha(X)$, a tabulated surface angle $\theta_\alpha(X)$, and a Froude number F_α . Xu and Guyenne likewise use a solitary wave computed by Tanaka's method [4]. The periodic embedding, tangent-Hermite interpolation, multi-profile superposition, and sampling laws below are choices of the present corpus rather than parts of either cited solitary-wave construction. The profile and angle satisfy

$$\frac{d\bar{\eta}_\alpha}{dX}(X) = \tan \theta_\alpha(X). \quad (22)$$

Before interpolation, the constructor checks that the two endpoint elevations and the endpoint and crest tangents differ from zero by at most 10^{-12} , and then sets those values exactly to zero. It forms a cubic Hermite interpolant from the corrected elevations and tangents and extends that interpolant by zero outside its tabulated interval. The value and first derivative therefore join continuously to the zero exterior. For a center x_0 , define its periodic embedding by

$$\eta_{\alpha,h,x_0}(x) = h \sum_{r \in \mathbb{Z}} \bar{\eta}_\alpha \left(\frac{x - x_0 + rL}{h} \right). \quad (23)$$

Because the extended profile has compact support, only finitely many terms are nonzero on one period. The implementation includes every translated copy whose support intersects $[0, L)$. Each copy is evaluated by the cubic Hermite interpolation just defined. The scaling is dimensionally consistent because

$$\partial_x \left[h \bar{\eta}_\alpha \left(\frac{x - x_0 + rL}{h} \right) \right] = \frac{d\bar{\eta}_\alpha}{dX} \left(\frac{x - x_0 + rL}{h} \right). \quad (24)$$

The sum is evaluated on an eight-times finer grid and then restricted to the native Fourier grid.

Set $c_\alpha = F_\alpha \sqrt{gh}$. For direction $s \in \{-1, 1\}$, the traveling-wave formula gives the candidate derivative

$$r_{\alpha,h,x_0,s} = s \left[c_\alpha - \sqrt{(1 + (\partial_x \eta_{\alpha,h,x_0})^2)(c_\alpha^2 - 2g\eta_{\alpha,h,x_0})} \right]. \quad (25)$$

Define its radicand by

$$\mathcal{R}_{\alpha,h,x_0}(x) = (1 + (\partial_x \eta_{\alpha,h,x_0}(x))^2) (c_\alpha^2 - 2g\eta_{\alpha,h,x_0}(x)). \quad (26)$$

The production constructor first requires c_α^2 to be finite and strictly positive and $\mathcal{R}_{\alpha,h,x_0}$ to be finite at every grid point. Failure of either requirement stops generation and is not relabeled as a support rejection. When these quantities are finite, a component with $\min_\ell \mathcal{R}_{\alpha,h,x_0}(x_\ell) < 0$ lies outside the declared Tanaka construction and is recorded as a zero-row case. The radicand is never modified by clipping.

Write $\widehat{r}_n$ for the n th Fourier coefficient of $r_{\alpha,h,x_0,s}$. A periodic antiderivative exists only after removing the mean, so the potential is defined by

$$\partial_x \xi_{\alpha,h,x_0,s} = P_0^\perp r_{\alpha,h,x_0,s}, \quad \widehat{\xi_{\alpha,h,x_0,s}}_n = \frac{\widehat{r}_n}{ik_n} \quad (n \neq 0), \quad \widehat{\xi_{\alpha,h,x_0,s}}_0 = 0. \quad (27)$$

Thus (25) is the unperiodized traveling-wave relation, while (27) is the actual periodic formula. Their difference is the constant $\widehat{r}_0$ and is recorded in the case audit.

For an integer component count $m \geq 1$, scaling, periodic embedding, and summation therefore give

$$(\eta_0, \xi_0) = P_{128} \sum_{j=1}^m (\eta_{\alpha_j, h, x_j}, \xi_{\alpha_j, h, x_j, s_j}). \quad (28)$$

Each Tanaka case records the depth h , component count m , and the amplitude ratio, center, and direction (α_j, x_j, s_j) of every component. The sharp projection P_{128} is applied once to the constructed initial state. For Tanaka revision 3, these arrays are then supplied unchanged to the $K_{\text{evol}} = 256$ solver. Equivalently, their Fourier coefficients for $129 \leq |k| \leq 256$ are zero at $t = 0$; those modes form an initially empty workspace that may be populated during the nonlinear evolution. Only the P_{128} state and label are exposed to the corpus.

Revision 3 samples amplitude before depth. In a main group with $m \in \{1, 2, 3\}$ crests, first draw the total dimensionless amplitude

$$A \sim \text{Unif}[0.10, 0.35]. \quad (29)$$

For $m > 1$, draw $(w_1^{(\alpha)}, \dots, w_m^{(\alpha)}) \sim \text{Dirichlet}(1, \dots, 1)$ and set $\alpha_j = Aw_j^{(\alpha)}$; for $m = 1$, set $\alpha_1 = A$. The steep group has $m = 1$ and instead draws

$$A = \alpha_1 \sim \text{Unif}[0.25, 0.45]. \quad (30)$$

In either group, define the largest component amplitude ratio by

$$\alpha_{\max} = \max_{1 \leq j \leq m} \alpha_j. \quad (31)$$

The small-amplitude KdV solitary-wave shape has the form $a \text{sech}^2(\kappa(x - x_0))$, with inverse-width scale

$$\kappa = \frac{\sqrt{3\alpha}}{2h}, \quad \alpha = \frac{a}{h}. \quad (32)$$

We use only this elementary length scale. It is not asserted to be the exact width of the finite-amplitude profile returned by Tanaka's method. The narrowest requested component has the proxy inverse width

$$\kappa_{\max} = \frac{\sqrt{3\alpha_{\max}}}{2h}. \quad (33)$$

Since the learned fields retain wavenumbers through 128, define the dimensionless resolution ratio

$$R = \frac{128}{\kappa_{\max}}. \quad (34)$$

Tanaka revision 3 requires $R \geq 10$. Let $(h_{\text{base}}, h_{\text{top}})$ equal $(0.01, 0.30)$ for a main group and $(0.20, 0.35)$ for the steep group. The inequality $R \geq 10$ is equivalent to

$$h \geq \frac{10\sqrt{3\alpha_{\max}}}{256}. \quad (35)$$

The conditional lower depth and sampled depth are therefore

$$h_{\min}(\alpha_{\max}) = \max\left\{h_{\text{base}}, \frac{10\sqrt{3\alpha_{\max}}}{256}\right\}, \quad h = h_{\min}(\alpha_{\max})^{1-U_h} h_{\text{top}}^{U_h}, \quad U_h \sim \text{Unif}[0, 1]. \quad (36)$$

Thus h is log-uniform on $[h_{\min}(\alpha_{\max}), h_{\text{top}}]$ after the amplitudes are known. For the steep group the resolution lower bound is always below 0.20, so its depth law remains log-uniform on $[0.20, 0.35]$.

For the main groups the rule removes only the shallow amplitude–depth combinations that do not meet the fixed P_{128} resolution condition. A deterministic audit with one million draws for each m found that the old independent-depth law placed 33.45%, 28.94%, and 25.81% of the $m = 1, 2, 3$ draws below $R = 10$. With the category weights below, the removed fraction is 28.55%. Under the new law, the median depths for $m = 1, 2, 3$ are respectively 0.0973, 0.0902, and 0.0857, and the depth support still reaches 0.30. These are checks of the sampling law, not claims about the natural frequency of water depths. Let

$$q = \#\{j : s_j = +1\} \quad (37)$$

be the number of right-moving crests. For the main groups, the pairs (m, q) with $m \in \{1, 2, 3\}$ and $q = 0, \dots, m$ define $2 + 3 + 4 = 9$ parameter categories. The steep group has the two categories $q = 0, 1$. Hence the Tanaka family has eleven parameter categories. Conditional on q , the sampler chooses uniformly which q of the m exchangeable crest labels have sign $+1$; all remaining signs are -1 . The accepted-case quota is divided equally over the eleven categories, up to one case when the requested total is not divisible by eleven. Thus left and right directions remain balanced marginally in the ideal equal-category population, but the direction compositions $q = 0, \dots, m$ have equal coverage instead of the binomial frequencies produced by independent fair signs. A finite quotient–remainder assignment may introduce a one-case direction imbalance; with the fixed category order, a quota of 1024 has one extra left-moving one-crest case. This law deliberately gives aggregate weights $2/11$, $3/11$, $4/11$, and $2/11$ to main $m = 1$, main $m = 2$, main $m = 3$, and steep $m = 1$, respectively. It replaces the old four-group independent-sign mixture. The direction categories were introduced in revision 2; revision 3 keeps those categories and changes the amplitude–depth order and conditional support just defined. Revision-2 Tanaka specifications remain exactly replayable, but they cannot be resumed as revision-3 generation.

The centers are sampled without privileging the endpoints of a plotting interval. Draw

$$(w_1^{(x)}, \dots, w_m^{(x)}) \sim \text{Dirichlet}(1, \dots, 1), \quad g_j = 3h + (L - 3mh)w_j^{(x)}, \quad (38)$$

use the g_j as the cyclic gaps between consecutive centers, and rotate all centers by one common uniform draw on $[0, L)$. Since $\sum_j g_j = L$, this defines a translation-invariant periodic configuration. The periodic distance $d_L(x_i, x_j) = \min_{r \in \mathbb{Z}} |x_i - x_j + rL|$ therefore satisfies

$$d_L(x_i, x_j) \geq 3h \quad (i \neq j). \quad (39)$$

For each α_j , the numerical profile solve returns one tabulated profile. Its tolerances and coefficient version are fixed for the entire dataset. Periodic continuation is part of the Tanaka construction, not a repair applied after sampling.

A sum of solitary profiles is a constructed initial state, not an exact multi-soliton solution. The eleven categories above belong to the same Tanaka family.

4.3 Benjamin–Feir carrier and sidebands

This family is deep-water only. Choose an integer carrier mode n_c , a positive integer sideband separation $\Delta n < n_c$, a carrier steepness ε_c , a ratio ρ equal to the amplitude of each sideband divided by the amplitude of the first carrier harmonic, and a global spatial translation $x_0 \in [0, L)$. Here ε_c is the steepness of the first elevation harmonic, not the wavenumber times half the total crest-to-trough height. Define

$$k_c = \frac{2\pi n_c}{L}, \quad k_{\pm} = \frac{2\pi(n_c \pm \Delta n)}{L}, \quad a = \frac{\varepsilon_c}{k_c}. \quad (40)$$

The sideband ansatz below follows equation (33) of Xu and Guyenne [4], which follows the high-order spectral construction of Dommermuth and Yue [7]. Their published calculation uses a numerically computed exact steady Stokes carrier. The present corpus instead uses the project's analytic fifth-order carrier and randomizes the parameters over the support stated below; it is not a reproduction of that single published case. Let $y = x - x_0$, and let $(\eta_c(y), \xi_c(y))$ be the project's analytic fifth-order deep-water Stokes carrier whose first elevation harmonic has amplitude a and whose phase is zero in the translated coordinate. This is an explicit approximation to the numerically exact steady Stokes carrier used by Xu and Guyenne; the two objects are not identified. The implemented initial state is the equation-(33) sideband form with the fixed relative sideband phase $\theta = -\pi/4$:

$$\begin{aligned}\eta_0 &= \eta_c(y) + \rho a [\cos(k_- y + \theta) + \cos(k_+ y + \theta)], \\ \xi_0(x) &= \xi_c(y) + \rho a \left[\sqrt{\frac{g}{k_-}} e^{k_- \eta_0(x)} \sin(k_- y + \theta) + \sqrt{\frac{g}{k_+}} e^{k_+ \eta_0(x)} \sin(k_+ y + \theta) \right],\end{aligned}\quad (41)$$

followed by removal of the spatial mean of ξ_0 . Both sidebands travel with the carrier. Translating the complete state changes neither its relative phases nor the quartet phase $\theta + \theta = -\pi/2$. Each case records $(n_c, \varepsilon_c, \Delta n, \rho, x_0, \theta)$, and the numerical depth is fixed to $h = 5L/(2\pi)$, so that even the first Fourier mode has $kh = 5$. There are no empirical cross terms and no exceptional rewrite of a sampled mode pair.

Define the normalized sideband location

$$\beta = \frac{\Delta n/n_c}{2\sqrt{2}\varepsilon_c}.\quad (42)$$

Benjamin and Feir give the historical deep-water instability condition [6]. In the present normalization, equation (4.6) of Andrade and Stuhlmeier gives $0 < \beta < 1$, while their equations (5.7)–(5.11) give the Akhmediev-breather correspondence and the focused-envelope amplification $1 + 2\sqrt{1 - \beta^2}$ [8]. Define the corresponding screening proxy

$$\varepsilon_f = \varepsilon_c \left(1 + 2\sqrt{1 - \beta^2} \right).\quad (43)$$

The declared upper bound is

$$F_0 = \frac{1 + \sqrt{2}}{10}.\quad (44)$$

This is the value of (43) at $\varepsilon_c = 0.10$ and the leading-NLS maximum-growth location $\beta = 1/\sqrt{2}$. It is a population-design choice anchored to a moderate recurrence calculation, not a wave-breaking threshold or a regularity theorem. Yang and Liu use $\varepsilon_c = 0.10$, five-percent sidebands, and the maximum-growth frequency coordinate $\Delta\omega/(\omega_c\varepsilon_c) = 1$ [9, Sec. 4.3.1]. Under the leading deep-water narrowband relation $\Delta k/k_c \simeq 2\Delta\omega/\omega_c$, this corresponds to $\beta \simeq 1/\sqrt{2}$ in our symmetric-wavenumber normalization. The ten-percent endpoint below is the sideband perturbation used by Xu and Guyenne [4].

The declared support is

$$\begin{aligned}n_c &\in \{4, \dots, 20\}, & \varepsilon_c &\in [0.05, 0.13], \\ 1 \leq \Delta n &< n_c, & 0 < \beta < 1, & \varepsilon_f \leq F_0, \\ \rho &\in [0.05, 0.10], & x_0 &\in [0, L],\end{aligned}\quad (45)$$

There are 66 feasible integer pairs $(n_c, \Delta n)$, and the focused bound leaves every one of them nonempty. For each family-split accepted-case quota, the production schedule assigns either $\lfloor C/66 \rfloor$ or $\lceil C/66 \rceil$

cases to each pair, where C is that quota; the first remainder categories in the frozen ordering receive the larger count. A rejected attempt is redrawn in the same pair category. Conditional on $(n_c, \Delta n)$, define

$$\ell(n_c, \Delta n) = \frac{\Delta n}{2\sqrt{2}n_c}, \quad \varepsilon_{\min}(n_c, \Delta n) = \max\{0.05, \ell(n_c, \Delta n)\}, \quad (46)$$

and

$$\varepsilon_{\max}(n_c, \Delta n) = \min\left\{0.13, \frac{-F_0 + 2\sqrt{F_0^2 + 3\ell(n_c, \Delta n)^2}}{3}\right\}. \quad (47)$$

The latter formula is obtained by solving $\varepsilon_f = F_0$ at fixed integer pair. Its value ranges from 0.081763 to 0.13 over the 66 pairs. The continuous draws are

$$\varepsilon_c \sim \text{Unif}(\varepsilon_{\min}, \varepsilon_{\max}), \quad \rho \sim \text{Unif}[0.05, 0.10], \quad x_0 \sim \text{Unif}[0, L], \quad \theta = -\frac{\pi}{4}. \quad (48)$$

These laws depend only on the sampled pair and not on a rollout outcome. The implemented population sampler treats the 66 pairs as parameter categories. Because the conditional steepness interval and sideband-ratio interval have changed, this population is Benjamin–Feir revision 4. It retains the revision-3 numerical evolution method, but a revision-3 proposal cannot be resumed into a revision-4 output root. At the final split quotas, the 16384 training cases give 248 or 249 accepted cases to every pair, while validation and test each give 15 or 16 of their 1024 accepted cases to every pair. Additive training chunks contain the differences between cumulative balanced allocations. The ordered adapter stores the realized pair and continuous parameters before calling the constructor.

The Xu–Guyenne diagnostic point is

$$n_c = 9, \quad \Delta n = 2, \quad \varepsilon_c = 0.13, \quad \rho = 0.1, \quad \theta = -\frac{\pi}{4}. \quad (49)$$

It is retained as a named, out-of-support diagnostic configuration [4]; it does not contribute training rows. Using linear sidebands in this construction does not create a separate Airy training family: the complete modulated state is one Benjamin–Feir case.

Define the linear carrier period and production horizon by

$$T_c = \frac{2\pi}{\sqrt{gk_c}}, \quad T_{\text{BF}} = 0.08 \left\lfloor \frac{100T_c}{0.08} \right\rfloor. \quad (50)$$

The intended horizon is $100T_c$, and the rollout ends at its strictly floored saved-grid endpoint T_{BF} . Across the 66 equally weighted integer-pair categories, T_{BF} ranges from 140.48 to 314.08 and has mean 174.50; equivalently, $T_{\text{BF}}/T_c \in [99.9541, 99.9973]$.

The support was suggested by retrospective calculations, so those calculations are not themselves a validation. In revision-3 stream 930, all 44 attempts satisfying $\varepsilon_f \leq F_0$ passed, but later proposals in that stream were triggered by earlier failures and are not an independent sample. In the older revision-2 archive, all 3,743 attempts satisfying the same focused-steepness inequality passed. That archive used a different evolution order, cutoff, phase law, and terminal-time convention, so its counts cannot be pooled with revision 3. They are only a cross-check that the same scalar restriction removes the old failure tail.

The support (45) was frozen before validation stream 934 was advanced. The complete parameter plan selected one independent case from each integer-pair category before numerical outcomes were known. All 66 cases were then evolved to the saved-grid endpoint immediately below their intended $100T_c$ horizon with the unchanged numerical method inherited by revision 4 from revision 3: the state retained Fourier modes $|k| \leq 256$, the internal Dirichlet–Neumann calculation used order four,

and the time step was 0.01. The calculation accepted all 66 attempts, rejected none, and made no outcome-driven replacement. All 13,200 retained frames were finite. The largest relative full-band Hamiltonian drift was 5.38713×10^{-5} , the largest implicit-stage residual was 9.999984×10^{-9} , and the smallest water column was 4.958158. Stream 934 therefore supplies independent support-validation evidence. It conditioned proposals from the preceding revision-3 law and is not the exact-source revision-4 production gate. That gate is stream 935, described in Section 5.

This calculation validates numerical usability of the sampled support under the stated discretization and horizon; it is not a proof about every point of the continuous support or about the full water-wave equations. In particular, ε_f is a leading-NLS focused-envelope proxy, not the maximum physical slope. The initial condition (41) is a finite Stokes–Airy state rather than an exact focused breather and therefore does not contain the breather’s full sideband ladder. Bound harmonics, finite sideband amplitude, and higher-order water-wave effects can change the realized focusing. The complete-trajectory finiteness, stage-residual, water-column, and Hamiltonian conditions in Section 6 therefore remain mandatory.

4.4 JONSWAP/TMA random-phase states

The random-sea family uses the JONSWAP frequency spectrum rather than the Gaussian spectrum in the historical **v9** generator. JONSWAP denotes the Joint North Sea Wave Project spectrum [11], and TMA denotes its standard Texel–Marsen–Arsloe finite-depth modification [12]. Choose a depth $h > 0$, peak wavenumber $k_p > 0$, and peak-enhancement factor $\gamma \geq 1$. For $k > 0$, define the finite-depth dispersion relation and group velocity by

$$\omega(k, h) = \sqrt{gk \tanh(kh)}, \quad c_g(k, h) = \frac{\partial \omega}{\partial k} = \frac{g[\tanh(kh) + kh \operatorname{sech}^2(kh)]}{2\omega(k, h)}. \quad (51)$$

Set the peak frequency $\omega_p = \omega(k_p, h)$ and define the peak period $T_p = 2\pi/\omega_p$. Let $k(\omega; h)$ denote the positive inverse of this dispersion relation. For angular frequency $\omega > 0$, define

$$\sigma(\omega) = \begin{cases} 0.07, & \omega \leq \omega_p, \\ 0.09, & \omega > \omega_p, \end{cases} \quad (52)$$

Define the unnormalized JONSWAP density directly by

$$\tilde{S}_J(\omega) = g^2 \omega^{-5} \exp\left[-\frac{5}{4} \left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^{\exp[-(\omega - \omega_p)^2 / (2\sigma(\omega)^2 \omega_p^2)]}. \quad (53)$$

For a dimensionless depth–wavenumber product ζ , define the TMA factor and finite-depth spectrum by

$$\Phi_{\text{TMA}}(\zeta) = \frac{\tanh^2 \zeta}{1 + 2\zeta / \sinh(2\zeta)}, \quad \tilde{S}_T(\omega; h) = \tilde{S}_J(\omega) \Phi_{\text{TMA}}(k(\omega; h)h). \quad (54)$$

This factor tends to one in deep water and behaves like $\zeta^2/2$ as $\zeta \rightarrow 0$ [12]. Merely using finite-depth dispersion in the η -to- ξ transfer, as the historical Gaussian-sea generator did, is not a TMA elevation spectrum. Revision 4 retains only the relative-frequency interval

$$\frac{1}{2}\omega_p \leq \omega \leq \frac{5}{2}\omega_p \quad (55)$$

and requires the upper endpoint to fit inside the delivered band:

$$\omega(128, h) \geq \frac{5}{2}\omega_p. \quad (56)$$

Hasselmann et al. supply the JONSWAP form [11], and Bouws et al. supply the TMA factor [12]. Draycott et al. use exactly the complete interval in (55) [14], while Grice et al. provide separate precedent for its upper endpoint $\omega_{\max} = 2.5\omega_p$ [13]. We adopt the interval as a finite component definition, not as a wave-breaking criterion. The Fourier-cell integration, direction split, phase sampling, and parameter strata below remain choices of the replacement corpus.

Let $\Delta k = 2\pi/L$ and use the positive-mode set $\mathcal{J}_K$ from (4). Thus K remains a physical wavenumber cutoff rather than a mode index. To avoid point-sampling a narrow spectral peak, integrate the continuous spectrum over Fourier cells. For each $j \in \mathcal{J}_K$, define its energy E_j using the clipped cell endpoints

$$a_j = \max\{0, k_j - \Delta k/2\}, \quad b_j = \min\{K, k_j + \Delta k/2\},$$

and

$$E_j = \int_{a_j}^{b_j} \tilde{S}_T(\omega(k, h); h) c_g(k, h) I_p(k) \, dk, \quad (57)$$

where

$$I_p(k) = \begin{cases} 1, & \frac{1}{2} \leq \frac{\omega(k, h)}{\omega_p} \leq \frac{5}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad (58)$$

The implementation evaluates each cell integral by fixed 16-point Gauss–Legendre quadrature, including the indicator at every quadrature node, so a boundary cell may be retained only partly. Define the normalized modal energy fractions

$$\varpi_j = \frac{E_j}{\sum_{\ell \in \mathcal{J}_K} E_\ell}, \quad \sum_{j \in \mathcal{J}_K} \varpi_j = 1. \quad (59)$$

The factor $c_g = \partial\omega/\partial k$ is the Jacobian from frequency density to wavenumber density. The parameter k_p is the wavenumber obtained by mapping the JONSWAP frequency-peak parameter through the finite-depth dispersion relation. TMA, cell integration, and I_p can shift the maximum of the discrete weights, so the realized maximizing mode is stored rather than assumed to equal k_p .

Let $H_s > 0$ denote the ensemble significant wave height and set $m_0 = (H_s/4)^2$. Let $r_d \in [0, 1]$ be the right-moving fraction in the traveling-wave decomposition of the linear Hamiltonian. For a positive scalar wavenumber, write $G_0(k; h) = k \tanh(kh)$ for the corresponding flat-surface DNO multiplier. Set

$$A_{j,+} = \sqrt{2r_d m_0 \varpi_j}, \quad A_{j,-} = \sqrt{2(1 - r_d) m_0 \varpi_j}. \quad (60)$$

For independent stored phases $\phi_{j,\pm}$, first define

$$\eta_0(x) = \sum_{j \in \mathcal{J}_K} [A_{j,+} \cos(k_j x + \phi_{j,+}) + A_{j,-} \cos(k_j x + \phi_{j,-})], \quad (61)$$

$$\xi_0(x) = \sum_{j \in \mathcal{J}_K} \frac{\omega(k_j, h)}{G_0(k_j; h)} [A_{j,+} \sin(k_j x + \phi_{j,+}) - A_{j,-} \sin(k_j x + \phi_{j,-})]. \quad (62)$$

Because right- and left-moving components can occupy the same spatial mode, the realized surface variance contains phase-dependent cross terms. Thus $H_s = 4\sqrt{m_0}$ is the spectral or ensemble significant height, not an exact identity for every realization. We do not rescale a realization to force that identity: when counterpropagating surface components nearly cancel, such a rescaling

can make the kinetic energy arbitrarily large. Define the linear Hamiltonian H_{lin} by the following phase-independent expression:

$$H_{\text{lin}} = \frac{1}{2} \int_{\mathbb{T}_L} (\xi_0 G_0(h) \xi_0 + g \eta_0^2) \, dx = g L m_0, \quad (63)$$

and r_d is exactly the right-moving fraction of this linear energy. The realized $\|P_0^\perp \eta_0\|_{2,L}$, $\|\xi_0\|_{2,L}$, and interference ratio $4 \|P_0^\perp \eta_0\|_{2,L} / H_s$ are recorded and included in the initial-geometry audit.

After the declared one-time spectral projection, a realization enters the rollout only if its two state fields are finite and

$$\min_{x_\ell} \{h + \eta_0(x_\ell)\} > 0. \quad (64)$$

This is the graph-domain condition that the free surface lie strictly above the flat bottom. If one realization in a numerical batch fails (64), only that realization becomes a zero-row attempt and is replaced in its own parameter category; valid batch companions are preserved. No realization-dependent amplitude cap is added.

Each valid case also records the realized discrete peak wavenumber, the number of spectral cells whose energy is at least half the peak-cell energy, the root mean squares of η_0 and ξ_0 , the ratio $4 \|P_0^\perp \eta_0\|_{2,L} / H_s$, the relative discrepancy in (63), and the minimum water column in (64). These quantities make the finite-grid construction auditable; none is an additional rejection threshold.

Let ϕ_+ and ϕ_- denote the arrays of right- and left-moving phases. Each random-sea case records $(h, H_s, k_p, \gamma, r_d, \phi_+, \phi_-)$. Storing the phase arrays is the most direct choice. A counter-based key is also sufficient only if the key-to-phase rule is versioned, indexed by physical mode, and stable across grid resolution.

The sharp indicator in (58) is tied to the prescribed peak frequency of each case; it is not a cosine taper at fixed absolute wavenumbers. Condition (56) ensures that the complete upper end of the interval lies inside the common delivered band $K = 128$. The resulting field is therefore a resolved-band, relative-frequency-truncated JONSWAP/TMA realization, not an exact untruncated TMA spectrum. This input-spectrum restriction is distinct from the common target projection P_K .

Increasing N at fixed L raises the largest resolved wavenumber but does not reduce the Fourier-cell width $2\pi/L$. The number of cells above half the largest cell energy is therefore recorded as a resolution diagnostic, not used to accept or reject a realization.

Sampling the dimensional variables independently is particularly dangerous in shallow water. Define instead

$$\chi_p = k_p h, \quad \delta_s = \frac{H_s}{2h}, \quad \varepsilon_p = \frac{k_p H_s}{2} = \chi_p \delta_s. \quad (65)$$

The quantity δ_s is a half-significant-height-to-depth descriptor; it is not the realized maximum crest height divided by depth. The corpus uses three strata of the same random-sea formula:

Table 1: JONSWAP/TMA training strata. The bounds define attempted cases; acceptance is reported separately.

Stratum	Attempted support
Shallow TMA	Reference-peak index n_p with $k_p = 2\pi n_p / L$ and $n_p \in \{16, 18, 20, 22, 24\}$, $\chi_p \in [0.2, 1.5]$, $h = \chi_p / k_p$, $\delta_s \in [0.03, 0.16]$, $\varepsilon_p = \chi_p \delta_s \leq 0.08$, and (56). For $L = 2\pi$, this gives $h \in [0.0083, 0.0938]$ and covers the diagnosed neighborhood $h \simeq 0.038$, $k_p h \simeq 0.7$.

Stratum	Attempted support
Finite-depth TMA	Draw $k_p \in [2, 12]$, $h \in [0.1, 1.5]$, and $H_s \in [0.005, 0.03]$, subject to $\varepsilon_p \leq 0.08$ and (56).
Deep JONSWAP	Draw $k_p \in [2, 12]$, $h \in [5, 25]$, and $H_s \in [0.005, 0.03]$, subject to $\varepsilon_p \leq 0.08$ and (56). Here the TMA factor is effectively one near the reference peak.

All three strata balance the peak-enhancement values $\gamma \in \{1, 3.3, 5\}$ and the right-moving fractions $r_d \in \{0, 1/2, 1\}$. Thus the production design has 27 parameter categories: one depth stratum, one value of γ , and one value of r_d . A requested family-split case count is divided among these categories by quotient and remainder, so their accepted-case quotas differ by at most one.

Within a shallow category, draw n_p uniformly from $\{16, 18, 20, 22, 24\}$. Draw χ_p uniformly on $[0.2, 1.5]$ and δ_s uniformly on $[0.03, 0.16]$, repeating this two-variable draw until $\chi_p \delta_s \leq 0.08$ and (56) both hold; equivalently, (χ_p, δ_s) is uniform with respect to area on the admissible part of that rectangle. Then set $k_p = 2\pi n_p / L$, $h = \chi_p / k_p$, and $H_s = 2h\delta_s$. Within a finite-depth or deep category, draw k_p , h , and H_s independently and uniformly on the three intervals in Table 1, repeating the triple until $k_p H_s / 2 \leq 0.08$ and (56) both hold. The resulting proposal is uniform on the portion of the corresponding rectangular box that obeys both bounds. In every category, all entries of the two phase arrays are independent and uniform on $[0, 2\pi)$. These proposal laws are implemented for all 27 categories.

The common bound $\varepsilon_p \leq 0.08$ declares a moderate input-sea population; it is not a breaking theorem. A JONSWAP/HOS study describes $k_p H_s / 2 = 0.08$ as typical of moderate sea states [16]. A separate applicability study shows that a coarse HOS cutoff can suppress short-wave content and make steep calculations appear more robust than a better-resolved calculation [15]. Because our potential-flow model contains no physical wave-breaking mechanism, we impose the moderate bound before phase sampling and still require the independent full-trajectory numerical test below.

All pseudorandom draws use deterministic, seeded streams produced by NumPy’s PCG64 generator. The durable proposal record stores the root seed, physical-family and revision identifiers, stream and attempt indices, every realized parameter, and both phase arrays. Before time integration, the common mathematical requirements are finiteness and the graph-domain condition $h + \eta_0(x) > 0$. The complete trajectory must still pass the common fixed-step numerical calculation in Section 6.

For JONSWAP/TMA, the event A_f in (68) specifically includes the phase-dependent initial construction and graph condition, followed by nonlinear adjustment and autonomous trajectory acceptance. This is not a separate phase-valued support gate: it is the common complete-case conditioning specialized to this family. The JONSWAP/TMA completion audit reports its effect separately; the common-law statement does not claim measured bias in another family. Every rejected proposal remains recorded, and no retrospective parameter cutoff or bulk rejection-rate gate is introduced.

A historical, pre-adjustment deterministic 27-case constructor pilot used nine cases in each stratum and every pair $(\gamma, r_d) \in \{1, 3.3, 5\} \times \{0, \frac{1}{2}, 1\}$ once per stratum. All 27 states were finite and in support. The largest relative error in the exact linear-energy identity (63) was 1.97×10^{-15} . The largest fixed-band $N = 512$ versus $N = 1024$ DNO discrepancy was 4.11×10^{-4} in shallow water, 7.76×10^{-5} at finite depth, and 6.07×10^{-5} in deep water. A short GL2-step check with $\delta = 0.01$ and 0.005 on the steepest member of each stratum had maximum discrepancy 3.85×10^{-8} . These are historical constructor and short-time checks, not evidence for the nonlinear adjustment, the autonomous full-horizon contract, or the revision-4 relative-frequency population. The constructor returns both realized phase arrays. The pilot field archive did not store those arrays; its JSON record stores a per-case seed from which the present implementation replays them. The production record

stores the arrays themselves, so replay does not depend on an unrecorded random-number call order.

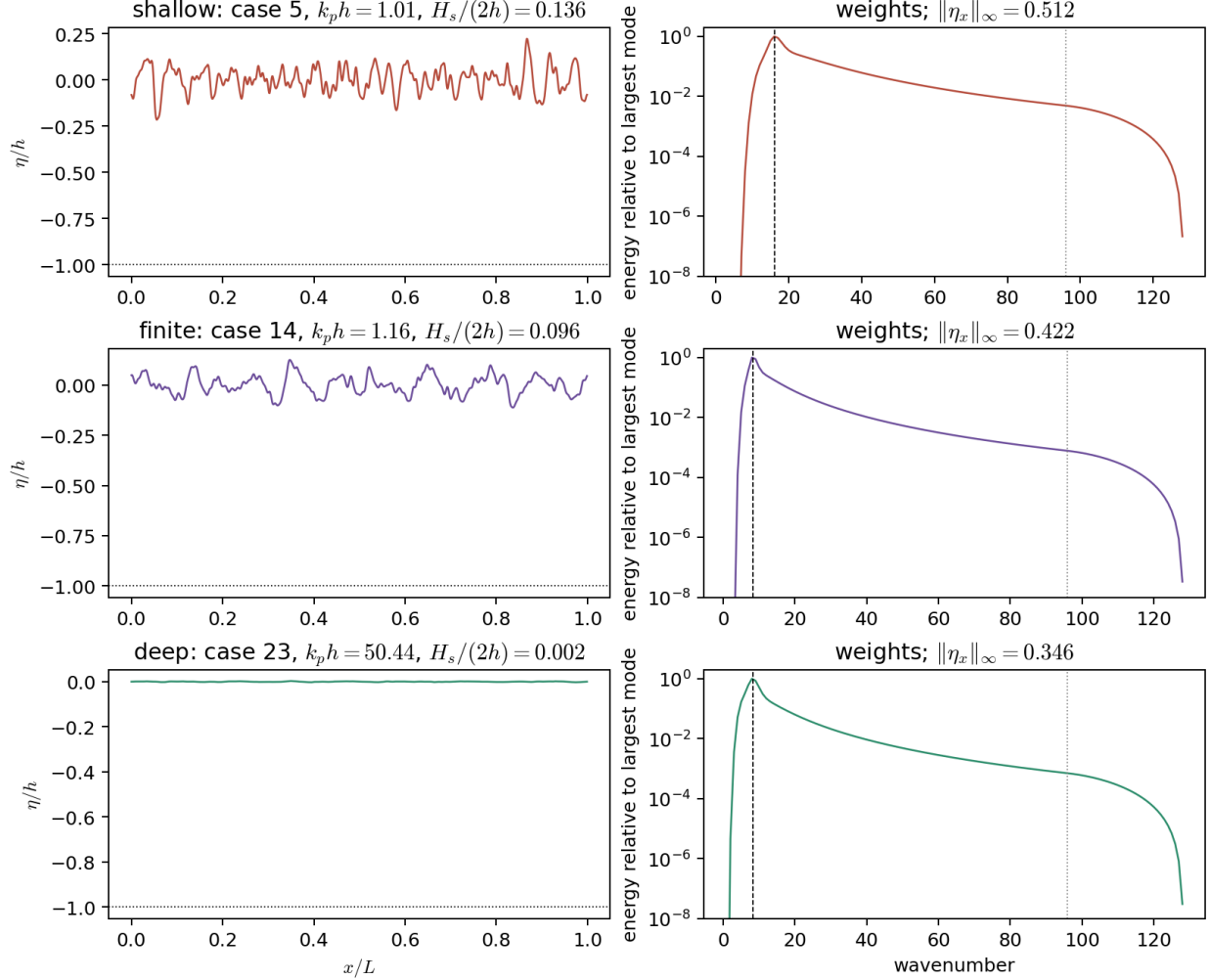


Figure 2: Representative stress cases from the historical revision-1, pre-adjustment, pre-relative-frequency JONSWAP/TMA constructor pilot. One case is shown from each depth stratum. These cases are selected by predeclared parameter stress and are not revision-4 corpus cases.

The shallow stratum replaces a role previously played by the first sparse packet addition. It does not automatically reproduce that data. Its roughness statistics and held-out shallow rollout accuracy are reported as diagnostics; neither is a production rejection rule. The design also does not silently claim the old packet minimum $h = 0.005$: extending below 0.0083 requires a separate resolved-peak and reference-convergence study.

Equations (61)–(62) define JONSWAP/TMA-shaped random-phase Zakharov states. Because ξ_0 uses the linear branch relation and contains no bound-wave correction, they should not be called fully nonlinear developed seas. Their role is controlled, physically interpretable DNO coverage.

4.5 Production parameter sampling

Each family first receives an accepted-case count C in each data split. Suppose the family has m ordered parameter categories and write

$$C = bm + r, \quad 0 \leq r < m. \quad (66)$$

The first r categories receive $b + 1$ accepted cases and the other categories receive b . Let $B_i(C)$ denote this cumulative quota for category i . An additive training chunk that begins at C_0 cases per family and adds A cases requests

$$B_i(C_0 + A) - B_i(C_0) \quad (67)$$

new accepted cases from category i . Balancing each chunk independently would repeatedly favor the same remainder categories and would not produce a balanced nested learning curve.

Each family paragraph states the continuous law inside each category. All four samplers use deterministic PCG64 streams. Before construction or numerical integration, each attempted case stores its split, parameter category, root seed, stream identifier, and realized parameters. If the attempt is rejected, its record remains in the manifest and the replacement is drawn from the same category.

The family formulas therefore describe attempted specifications, not the unconditional law of loader-visible cases. If X is a complete attempted specification in preassigned category i , A_f is the declared complete case-acceptance event for family f , and w_i is the accepted-quota weight, then

$$p_{\text{release}}(i, x) = w_i p_{\text{proposal}}(x \mid i, A_f = 1). \quad (68)$$

For Stokes, A_f comprises declared static support, a finite state, positive water column, and a finite target. For Tanaka, it comprises declared construction support followed by complete autonomous-trajectory acceptance; for Benjamin–Feir, initial-state validity followed by complete autonomous-trajectory acceptance. For JONSWAP/TMA, it comprises initial construction and graph validity followed by nonlinear adjustment and complete autonomous-trajectory acceptance. Thus accepted quotas preserve declared category marginals, not the unconditioned proposal law within a category. This identity states how the release is constructed; it does not by itself claim a measured within-category shift for every family. No retrospective parameter cutoff is applied.

The replacement loop is finite. If category i requests $Q_i > 0$ accepted cases, it permits at most

$$A_i^{\max} = 4Q_i \quad (69)$$

durable case attempts. One attempt is one assigned case, not an internal Stokes amplitude redraw or a production GL2 rollout. A proposal reserves its attempt once, even if execution is interrupted. Pending work at the ceiling is replayed. If all $A_i^{\max}$ attempts have resolved and fewer than Q_i cases were accepted, the family run terminates and cannot enter a combined dataset. Thus the factor four is a computational bound, not a realized-field rejection rule. When the quota completes before the bound, it does not change the sampled or accepted cases.

4.6 Durable generation and dataset assembly

Each attempt is written in three possible stages:

proposal record $\longrightarrow$ accepted field file, when applicable $\longrightarrow$ result record.

The proposal is written first and contains the split, parameter category, random-stream coordinates, and complete sampled specification. The result record is the commit marker. If execution stops

after the proposal or after the field file, restarting replays the same attempt rather than drawing a replacement. A result that names a missing field file, or a file with a different SHA-256 digest, is treated as corruption.

The run identity binds the requested quota, family, split, batch size, platform, random stream, numerical settings, software versions, dependency files, and generator source files. A change in any of these items cannot silently resume the same output directory. The combined dataset builder checks every included record and file, requires complete and gap-free nested chunks with equal accepted family counts in each split, and writes a compact row-to-case index. It does not copy the field arrays. One dataset view contains exactly one revision of each physical family. Numerical execution and generator-source identities are compared within each (family, revision) pair. They are not required to be identical between different families, because the Tanaka revision-3 buffer is an intentional family-specific numerical contract. The delivered DNO target, stored dtypes, whole-case row rule, and dependency environment remain common view-level requirements.

4.7 Training corpus at a glance

Table 2 summarizes the replacement corpus. It assigns the same accepted-case count to each family and balances the listed coarse parameter ranges within each family.

Table 2: Completed four-family training corpus. “Times” counts stored states from a complete admissible fixed-step trajectory.

Family	Balanced parameter categories	Reference horizon and stored times	Case weight
Stokes	Finite/deep analytic branches; low and moderate categories from (16) within the current amplitude support, with finite-depth draws satisfying (20).	Static; one state per case.	1/4
Tanaka	Eleven (regime, m , q) categories, where q is the number of right-moving crests; amplitudes are drawn first, depth then satisfies $R \geq 10$, and amplitudes, centers, and separations are recorded explicitly.	Final time 200; 200 times selected from the accepted trajectory using the surface-gradient activity in (104).	1/4
Benjamin–Feir	Deep-water carrier mode and steepness, one symmetric sideband separation, one sideband ratio, and one global translation from (45)–(42).	Last 0.08-grid time not exceeding $100T_c$; 200 approximately uniform times including both endpoints.	1/4
JONSWAP/TMA	Shallow, finite, and deep strata from Table 1; γ , direction fraction, peak depth, relative height, and phase categories balanced explicitly.	First adjust to the last 0.08-grid time no later than $20T_p$; then restart the autonomous clock. Reference horizon $H = 16(2\pi/\omega_p)$; realized endpoint $T_s = .08\lfloor H/.08 \rfloor \leq H$; 16 approximately uniform saved autonomous times including both endpoints.	1/4

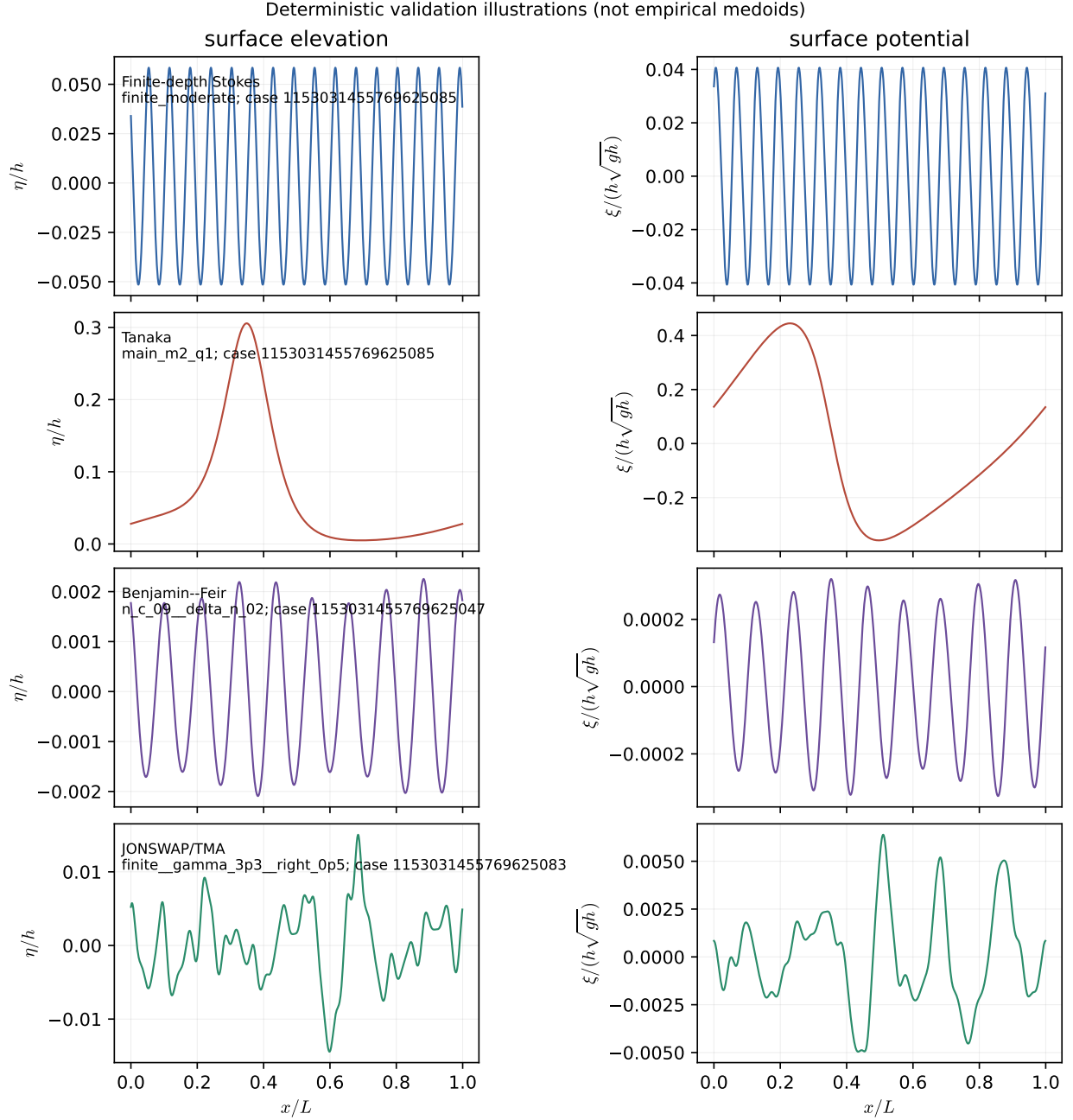


Figure 3: Deterministic illustrations from accepted validation cases in one declared central category per family, selected by the lower-median case ID; they are not empirical medoids. The rows are Stokes revision 2 case 1153031455769625085 (`finite_moderate`), Tanaka revision 3 case 1153031455769625085 (`main_m2_q1`), Benjamin-Feir revision 4 case 1153031455769625047 (`n_c_09_delta_n_02`), and JONSWAP/TMA revision 4 case 1153031455769625083 (`finite_gamma_3p3_right_0p5`). Each panel shows the first retained row. The columns are the dimensionless elevation η/h and surface potential $\xi/(h\sqrt{gh})$, with horizontal coordinate x/L .

Within the JONSWAP/TMA family the design gives equal case weight to its three depth strata. Within the other families the listed coarse strata are likewise balanced before seeded pseudorandom sampling within each category. In particular, the eleven Tanaka direction-count categories are balanced as evenly as each finite quota permits, so their counts differ by at most one. This gives additional aggregate weight to two- and three-crest states because they have more direction compositions. This is a transparent coverage design, not a claim that all categories are equally common in nature. If C is the accepted training-case count per family selected by the learning curve, the four-family training archive contains $C + 200C + 200C + 16C = 417C$ stored states before any storage-level compression. Thus the planned learning-curve grid $C \in \{2048, 4096, 8192, 16384\}$ corresponds to 854,016 through 6,832,128 training rows, while retaining $4C$ accepted training-case units. Validation, test, and rejected attempted cases are counted separately. The first checkpoint uses $C = 2048$ training cases per family and 1024 validation and 1024 test cases per family. It contains exactly $417(2048 + 1024 + 1024) = 1,708,032$ retained rows. The production target is $C = 16384$, with validation and test unchanged. Its 7,686,144 rows are close to the 7,633,989 rows in historical v9: 52,155 more rows, or 0.68%. The authenticated final view references 1,315 physical field shards containing 94,666,557,624 measured bytes (88.165102 GiB). Each smaller cumulative view references an immutable source prefix rather than copying fields. Thus $C = 2048$ is a cumulative learning-curve checkpoint, not the completed replacement corpus. Rejected cases are not replaced by copying nearby accepted cases; the attempted and accepted counts are both reported. Using a fixed number of peak periods prevents shallow categories from facing a much longer acceptance test merely because their dimensional time scale differs. The historical packet *training generator* stored trajectories to final time 120. That archive does not set either the new packet diagnostic horizon of 20 time units or the random-sea training horizon.

5 Attempted and accepted counts

The tables above specify attempted populations. For every family and every declared parameter category, report both attempted and accepted counts. A category with no accepted cases is not filled from another category. Instead, generation stops and the support or numerical method must be revised explicitly.

The population checks are computational. The complementary Tanaka revision-3 diagnostic chunks completed with one accepted case in each of their eleven categories and no rejection. Benjamin–Feir revision-4 stream 935 likewise tested one direct production-law proposal from each of its 66 categories. The current JONSWAP/TMA constructor instead used the larger predeclared 540-case revision-4 gate described below.

The Benjamin–Feir exact-source production gate, revision-4 stream 935, accepted all 66 first proposals, rejected none, and retained 13,200 rows. Its maximum implicit stage residual was 9.999967×10^{-9} , maximum relative internal Hamiltonian drift was 2.952009×10^{-5} , and minimum water column was 4.960598.

The preceding JONSWAP/TMA revision-3 stream 933 is retained only as historical evidence for the inherited adjustment and evolution method. It accepted one case in every category after 29 attempts and retained 432 rows. Its first proposals in `finite__gamma_3p3__right_1` and `shallow__gamma_1__right_0p5` failed during nonlinear adjustment and retained zero rows; each canonical second proposal passed. Over the 27 accepted cases, the maximum residual was 9.999783×10^{-9} , maximum Hamiltonian drift was 5.908800×10^{-5} , and minimum water column was 0.0127322. Thus its exact historical count was 27/29.

The independent audit rederived all saved proposals and run specifications and passed at

outputs/noncorpus_bf_revision4_jonswap_revision3_category_smoke_20260802/category_smoke_release_audit_v1.json. Its SHA-256 is 62fb5fe3b6be7df0d21784b1ce590c44cc484bf5fdf84ad6097fbed6f655f684. This artifact independently audits the Benjamin–Feir revision-4 gate and the historical JONSWAP/TMA revision-3 gate; it does not validate the revision-4 relative-frequency constructor. The earlier Benjamin–Feir stream-934 result remains support-validation evidence, while stream 935 is its exact-source revision-4 gate. Benjamin–Feir production is complete under outputs/paper_corpus_bf_revision4_jonswap_revision3_literature_aligned_v1. Only its Benjamin–Feir revision-4 summaries enter the final corpus; any JONSWAP/TMA revision-3 artifact under that historically named root is excluded. The full source-bound completion audit is outputs/paper_corpus_bf_revision4_jonswap_revision3_literature_aligned_v1/benjamin_feir_completion_audit.json, with SHA-256 c155b35276d0cfef6844f6b0db3e91da3ce15579ad8b0f0ca482f44cc912753c. It independently verifies the current ordered 66-category taxonomy and split quotas, reconstructed attempted/accepted/rejected counts in every category, exact proposal replay, transaction and array hashes, accepted and rejected row ownership, the 200 endpoint-pinned, approximately uniform selections from each saved grid, stored depths and water columns, and the declared residual and Hamiltonian conditions. Two shared Python modules changed after the run and their exact run versions were absent from the Git history. Their bytes are preserved as an inert audit copy, together with a matching SHA256SUMS file, under reproducibility/source_snapshots/, in the subdirectory benjamin_feir_revision4_e16773f/. This directory is not a current import path. The full-corpus audit remains the binding record for the arrays and transactions, and physically rehashes the other 17 recorded generation sources against the current repository. It is distinct from the one-case-per-category gate above.

The predeclared JONSWAP/TMA revision-4 gate fixed 20 accepted cases in each of the 27 categories before execution. It retained 540 cases from 548 attempts; eight incomplete numerical attempts retained zero rows. The observed rejection rate was 1.46%, below the predeclared two-percent bound. This was the completed constructor-adoption criterion, not a bulk release threshold. Its audited summary is outputs/jonswap_relative_band_fresh_gate_20260804/jonswap_relative_band_fresh_gate.summary.json. Revision-4 bulk generation uses outputs/paper_corpus_jonswap_revision4_relative_band_v1. The random-sea 2048-point $K = 704$ versus $K = 768$ sentinel remains the separate global whole-field cutoff comparison described below.

6 Reference evolution and whole-case acceptance

6.1 Discrete Zakharov flow

The reference trajectory is the solution of a declared discrete vector field, not merely the output of the two-stage, fourth-order Gauss–Legendre method (GL2). The depth h remains fixed. Let K_{evol} denote the largest wavenumber retained while a trajectory is evolved. For $w = (\eta, \xi)$, let M_F be the evolution order assigned to family F , and define the internal DNO evaluation and the evolution-band output by

$$G_{\text{int}, K_{\text{evol}}}(\eta; h)\xi = \mathcal{G}_{M_F}^{N, p}(P_{K_{\text{evol}}}\eta; h)(P_{K_{\text{evol}}}\xi), \quad G_{\text{evol}}(\eta; h)\xi = P_{K_{\text{evol}}}^{\circ}[G_{\text{int}, K_{\text{evol}}}(\eta; h)\xi]. \quad (70)$$

The nonlinear product is formed with the unprojected internal output before the evolution-band projection. The discrete vector field is

$$f_{h,K_{\text{evol}}}(w) = \begin{bmatrix} G_{\text{evol}}(\eta; h)\xi \\ P_{K_{\text{evol}}}^\circ \left[-g\eta - \frac{1}{2}\xi_x^2 + \frac{(G_{\text{int},K_{\text{evol}}}(\eta; h)\xi + \eta_x\xi_x)^2}{2(1+\eta_x^2)} \right] \end{bmatrix}. \quad (71)$$

Let $w_0 = (\eta_0, \xi_0)$ be an admissible candidate initial state, and write $z_0 = (\eta_0, \xi_0, h)$ when the fixed depth is also needed. The reference ODE is

$$\frac{dw}{dt} = f_{h,K_{\text{evol}}}(w), \quad w(0) = w_0. \quad (72)$$

The products in the second component are evaluated on the declared dealiased grid before projection. We integrate this ODE with the GL2 scheme in the next subsection. Removing the mean of $\partial_t \xi$ is precisely the time-dependent additive-potential gauge; it preserves $\int_{\mathbb{T}_L} \xi \, dx = 0$ without changing the velocity field or the surface evolution. The constructor sharply projects both initial fields to $|k| \leq 128$. For Tanaka and Benjamin–Feir, this P_{128} state is embedded without alteration in the 1024-point, $K_{\text{evol}} = 256$ evolution space. For JONSWAP/TMA it is embedded in the 2048-point, $K_{\text{evol}} = 704$ space. Thus every coefficient above mode 128 initially vanishes. The JONSWAP/TMA adjustment defined below subsequently replaces this zero-filled state by its full-band endpoint. Inside every right-hand-side evaluation, both nonlinear components are sharply projected to the relevant evolution band, with the mean additionally removed from the ξ component as in (5). These projections are part of $f_{h,K_{\text{evol}}}$.

The evolution contract depends on the family and its generator revision:

Family and revision	N	M_F	K_{evol}	Completed-step state map	Delivered band
Tanaka revision 3	1024	6	256	power-36 Hou–Li	128
Benjamin–Feir revision 4	1024	4	256	sharp P_{256}	128
JONSWAP/TMA revision 4	2048	4	704	sharp P_{704}	128

For Tanaka revision 3, after every completed GL2 step the Fourier coefficient of each state field is multiplied by

$$\sigma_k = \exp \left[-36 \left(\frac{|k|}{256} \right)^{36} \right]. \quad (73)$$

The same multiplier is applied to η and ξ . Xu and Guyenne use the power-36 Hou–Li form in their numerical method [4]; the internal cutoff 256 is selected here by the project-specific resolution comparison below. For all families, the arrays presented to the corpus are $P_{128}\eta$, $P_{128}\xi$, and the common 1024-point order-six target $G_{\text{ref}}(\eta; h)\xi$ from (11), evaluated from those delivered fields. Modes above 128 in every trajectory rollout are numerical workspace; they are neither network inputs nor labels. For Benjamin–Feir and JONSWAP/TMA, $M_F = 4$ determines the internal dynamics and $M = 6$ defines the delivered learning target.

For the integrating factor, define the linear part L_h and the remainder $r_{h,K_{\text{evol}}}$ by

$$L_h(\eta, \xi) = (G_0(h)\xi, -gP_0^\perp \eta), \quad r_{h,K_{\text{evol}}}(w) = f_{h,K_{\text{evol}}}(w) - L_h w. \quad (74)$$

Benjamin–Feir starts the autonomous equation (72) directly. The revision-4 JONSWAP/TMA path first uses the Dommermuth nonlinear adjustment inherited from revision 3 [10]

$$\dot{w} = L_h w + A(t)r_{h,704}(w), \quad A(t) = 1 - \exp \left[- \left(\frac{t}{10T_p} \right)^4 \right]. \quad (75)$$

Dommermuth supplies the ramped-nonlinearity construction; the exponent four, the scale $10T_p$, and the $20T_p$ handoff horizon are fixed project settings validated below rather than unique prescriptions of that method. For each random-sea case, this nonautonomous calculation stops at

$$T_{\text{adj}} = 0.08 \left\lfloor \frac{20T_p}{0.08} \right\rfloor, \quad (76)$$

the last saved-grid time not exceeding $20T_p$. Its complete 2048-point, $K_{\text{evol}} = 704$ endpoint is handed to the autonomous solver without projection to P_{128} and without rescaling either component. The autonomous clock is restarted at zero, A is fixed to one, and its intended production horizon is $16T_p$; the realized endpoint is the last saved-grid time $T_s \leq 16T_p$. The new Hamiltonian reference is the handoff state. The Hamiltonian-conservation condition below is not imposed during adjustment because (75) depends explicitly on time. Finiteness, positive water depth, and convergence of every implicit stage are still required during this adjustment.

On each nonzero Fourier mode, define $\omega_n^2 = g|k_n| \tanh(h|k_n|)$. The linear evolution is the explicit rotation

$$\begin{bmatrix} \widehat{\eta}_n(t) \\ \widehat{\xi}_n(t) \end{bmatrix} = \begin{bmatrix} \cos(\omega_n t) & \frac{|k_n| \tanh(h|k_n|)}{\omega_n} \sin(\omega_n t) \\ -\frac{g}{\omega_n} \sin(\omega_n t) & \cos(\omega_n t) \end{bmatrix} \begin{bmatrix} \widehat{\eta}_n(0) \\ \widehat{\xi}_n(0) \end{bmatrix}. \quad (77)$$

The zero modes are unchanged. We write this linear evolution as e^{tL_h} . With $v(t) = e^{-tL_h}w(t)$,

$$\dot{v} = \widetilde{F}(t, v) := e^{-tL_h} r_{h, K_{\text{evol}}}(e^{tL_h} v). \quad (78)$$

During the JONSWAP/TMA adjustment, the right-hand side in (78) is multiplied by $A(t)$.

6.2 Residual-controlled Gauss–Legendre stages

The two-stage, fourth-order Gauss–Legendre coefficients are

$$c_{1,2} = \frac{1}{2} \mp \frac{\sqrt{3}}{6}, \quad A = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \end{bmatrix}. \quad (79)$$

Let t_n and v_n be the current time and integrating-factor state, and let $\delta_n > 0$ be the step length. Write a_{ij} for the entries of the matrix A in (79). The two stage values satisfy

$$V_i = v_n + \delta_n \sum_{j=1}^2 a_{ij} \widetilde{F}(t_n + c_j \delta_n, V_j), \quad i = 1, 2. \quad (80)$$

After the stages converge,

$$v_{n+1} = v_n + \frac{\delta_n}{2} \sum_{i=1}^2 \widetilde{F}(t_n + c_i \delta_n, V_i). \quad (81)$$

For a candidate pair (V_1, V_2) , define the fixed-point maps

$$\Phi_i(V_1, V_2) = v_n + \delta_n \sum_{j=1}^2 a_{ij} \widetilde{F}(t_n + c_j \delta_n, V_j), \quad i = 1, 2. \quad (82)$$

Write V_i^η, V_i^ξ and Φ_i^η, Φ_i^ξ for their two Fourier-space components, and let $\|\cdot\|_{\ell^2}$ be the Euclidean norm of one component's Fourier coefficients. The implemented relative stage defect is

$$R_n = \max_{\substack{i \in \{1,2\} \\ u \in \{\eta, \xi\}}} \frac{\|\Phi_i^u - V_i^u\|_{\ell^2}}{\max\{\|\Phi_i^u\|_{\ell^2}, \|V_i^u\|_{\ell^2}, \text{tiny}\}}, \quad (83)$$

where tiny is the smallest positive normal number of the working floating-point type. Normalizing η and ξ separately avoids adding fields with different units. By Parseval's identity, the Fourier normalization cancels from each ratio.

The stage tolerance τ_{GL} and iteration cap are fixed in the generator specification. A stage is declared converged only when $R_n \leq \tau_{\text{GL}}$; otherwise its iteration ends at the declared cap or when a nonfinite stage is detected. Xu and Guyenne report that two to four iterations typically achieved relative errors between 10^{-6} and 10^{-8} in their applications [4]; that observation motivates a tolerance study, not an unchecked iteration count. The implementation recomputes (83) from the returned stages, freezes each batched sample after convergence, and records finite cap exhaustion separately from a nonfinite stage or state. With $\tau_{\text{GL}} = 10^{-8}$ and an eight-iteration cap, a four-code-path $N = 256$ smoke converged all 32 sampled steps in two updates. The largest residuals for Airy, finite Stokes, Tanaka, and the pre-revision BF constructor were respectively 9.08×10^{-11} , 1.10×10^{-10} , 4.63×10^{-10} , and 2.09×10^{-9} . That eight-update calculation is historical smoke evidence. The current paper-corpus contract permits at most four fixed-point updates per GL2 stage for Tanaka and Benjamin–Feir and at most five for JONSWAP/TMA. Every family requires the same 10^{-8} residual.

6.3 Method-level fixed-band resolution validation

This validation is performed on a predeclared panel before production and does not accept or reject individual corpus cases. Resolution is tested by reconstructing one physical specification, not by interpolating one sampled array. Fix a complete case record and re-evaluate the same family formula with the same stored parameters on the three grids $N_r = 2^r N$, $r = 0, 1, 2$. Each reconstruction is performed independently in float64. Write η_r and ξ_r for the fields reconstructed on grid N_r . Define the dimensionless fields

$$U_r^{(\eta)} = \frac{\eta_r}{h}, \quad U_r^{(\xi)} = \frac{\xi_r}{h\sqrt{gh}},$$

and write $\widehat{U}_{r,n}^{(\nu)}$ for the Fourier coefficient of $U_r^{(\nu)}$, where $\nu \in \{\eta, \xi\}$. The initial-state discrepancy is

$$\Delta_{\text{IC},K} = \max_{\nu \in \{\eta, \xi\}} \max_{0 \leq r < s \leq 2} \frac{\left(\sum_{|k_n| \leq K} |\widehat{U}_{r,n}^{(\nu)} - \widehat{U}_{s,n}^{(\nu)}|^2 \right)^{1/2}}{\left(\sum_{|k_n| \leq K} |\widehat{U}_{s,n}^{(\nu)}|^2 \right)^{1/2} + 10^{-30}}. \quad (84)$$

On grid N_r , compute the target field

$$G_r^* = P_K^\circ \left[\mathcal{G}_M^{N_r, p}(P_K \eta_r; h)(P_K \xi_r) \right], \quad (85)$$

set $U_r^{(G)} = G_r^* / \sqrt{gh}$, and write $\widehat{U}_{r,n}^{(G)}$ for its Fourier coefficients. Then

$$\Delta_{G,K} = \max_{0 \leq r < s \leq 2} \frac{\left(\sum_{|k_n| \leq K} |\widehat{U}_{r,n}^{(G)} - \widehat{U}_{s,n}^{(G)}|^2 \right)^{1/2}}{\left(\sum_{|k_n| \leq K} |\widehat{U}_{s,n}^{(G)}|^2 \right)^{1/2} + 10^{-30}}. \quad (86)$$

The frozen DNO threshold for this Tanaka configuration-validation panel is $\Delta_{G,K} \leq 10^{-3}$. Both statistics are evaluated on its predeclared cases before the Tanaka construction is frozen. Other families use the separately stated spatial or cutoff calculations; evidence from one family or revision is not silently promoted to another. A failed panel identifies an unresolved constructor or unsupported parameter category; it does not authorize deleting individual production rows by inspecting their shape. Thus the expensive order- M calculation is not repeated on three grids at every stored time.

The historical Tanaka generator instead used a rowwise sign-transition rule. If $D_j = [G_{\text{old}}(\eta; h)\xi](x_j)$ is the sampled old target, that rule formed the cyclic forward difference $d_j = (D_{j+1} - D_j)/\Delta x$, discarded entries below $0.03 \max_j |d_j|$, and rejected when the remaining cyclic sign sequence had more than ten transitions. It was useful for the old clipped-profile constructor because a boundary seam generated oscillatory curvature, but a resolved physical wave may also have many extrema. In the completed Tanaka calibration at the historical band $K = 341$, all 12 matched clipped cases had $\Delta_{G,K} \geq 2.33 \times 10^{-2}$; the same specifications rebuilt by the periodic constructor had $\Delta_{G,K} \leq 1.10 \times 10^{-5}$, and four clean controls had $\Delta_{G,K} \leq 1.64 \times 10^{-4}$. This supports the mechanism without claiming that the current corpus was already filtered by the three-grid statistic. The tangent-Hermite construction removes the seam mechanism, the three-grid panel checks its spatial resolution before production, and production applies no sign-transition rejection.

The revision-3 resolution support was chosen by a separate authenticated full-horizon calculation. Nine one-crest specifications crossed $\alpha \in \{0.225, 0.30, 0.35\}$ with $R \in \{6, 8, 10\}$, where R is defined in (34). Each specification was reconstructed once, then evolved through $T = 200$ with otherwise matched internal cutoffs 224 and 256. Both calculations used float64, the multiplier (73) with its denominator changed to the corresponding internal cutoff, 2,501 saved frames, and 20,000 GL2 steps. Every implicit stage was solved in all eighteen trajectories.

All three $R = 6$ controls failed the predeclared 10^{-3} comparison. Their largest discrepancy in the delivered $G(\eta; h)\xi$ was 9.36×10^{-3} . Write $\eta_K(x, t)$ for the delivered P_{128} surface obtained with internal cutoff $K \in \{224, 256\}$. To measure the part associated with translation, at each saved time t we chose the continuous shift $\delta(t)$ that maximized the discrete cross-correlation between $\eta_{224}(\cdot, t)$ and $\eta_{256}(\cdot, t)$. For

$$\text{rms}_x[v] = \left(\frac{1}{N} \sum_{j=0}^{N-1} |v(x_j)|^2 \right)^{1/2},$$

the reported translation-effect statistic was

$$E_{\text{shift}} = \frac{\max_t \text{rms}_x \left[\frac{\eta_{256}(\cdot + \delta(t), t) - \eta_{256}(\cdot, t)}{h} \right]}{\max_t \text{rms}_x \left[\frac{\eta_{256}(\cdot, t)}{h} \right] + 10^{-12}}. \quad (87)$$

The largest E_{shift} among the $R = 6$ controls was 4.02×10^{-3} , and their largest hidden-band closure defect was 2.91×10^{-3} . All $R = 8$ and $R = 10$ cases passed. At $R = 8$, the largest discrepancies over the three amplitudes were

$$1.87 \times 10^{-4} \quad \text{in } \eta, \quad 1.29 \times 10^{-5} \quad \text{in } \xi, \quad 7.34 \times 10^{-4} \quad \text{in } G(\eta; h)\xi, \quad (88)$$

and the largest hidden-band closure defect was 6.31×10^{-4} . Thus $R = 8$ passed but used most of the 10^{-3} budget. At $R = 10$, the corresponding values fell to 2.94×10^{-5} , 1.62×10^{-6} , 1.35×10^{-4} , and 1.40×10^{-4} . This is a factor of about 7.1 below the budget and motivates the round support value $R \geq 10$.

The two internal cutoffs have the same $N = 1024$ physical arrays and transform sizes in this implementation. The exact two-times buffer $K_{\text{evol}} = 256$ is therefore retained instead of 224. The largest recorded Hamiltonian drift among the $R = 10$ calculations was 2.99×10^{-8} , but that value is only corroborating evidence from this fixed validation panel. It is not a production acceptance condition. The boundary calculation supports the conditional parameter law and the internal resolution; it does not establish that the KdV proxy is an exact Tanaka width.

6.4 Method-level selection of the production time step

Two different intervals were called $\mathbf{dt}$ in earlier project notes. Let Δs denote the spacing between saved states and let δ denote the actual GL2 step in (80)–(81). If one saved interval is divided into m equal GL2 steps, then

$$\delta = \frac{\Delta s}{m}. \quad (89)$$

Production uses

$$\Delta s = 0.08, \quad m = 8, \quad \delta = 0.01. \quad (90)$$

Thus 0.08 is a storage interval, not the step of the GL2 method.

The choice $\delta = 0.01$ was made before the present corpus plan. The May 2026 time-step study used $m = 8$ while varying the saved interval through $\Delta s \in \{0.01, 0.02, 0.04, 0.08, 0.16, 0.32\}$. In terms of the actual GL2 step, the tested values were therefore

$$\delta \in \{0.00125, 0.0025, 0.005, 0.01, 0.02, 0.04\}.$$

Four Tanaka cases were integrated to $T = 50$. Relative terminal elevation differences from the $\delta = 0.00125$ calculation were about 10^{-11} , 10^{-10} , and 10^{-9} at $\delta = 0.0025$, 0.005 , and 0.01 , respectively. The difference jumped to 6.2×10^{-2} at $\delta = 0.02$. The historical statement “use $\mathbf{dt}=0.08$ ” therefore selected the largest successful *saved interval with eight substeps*; it selected the actual GL2 step $\delta = 0.01$, not $\delta = 0.08$.

Xu and Guyenne use the same GL2 step 0.01 in their reported calculations [4]. Their paper does not determine our saved spacing, residual tolerance, or complete-case policy. Those are numerical choices for this corpus.

The step choice was checked again in July 2026 on twelve predeclared full-horizon cases from the trajectory families. For this method-level validation only, each initial condition was integrated with $\delta = 0.01$ and 0.005 and compared at common saved times. At a saved time t_j , define

$$Y_{\delta}^{(\eta)}(t_j) = \frac{P_K \eta_{\delta}(t_j)}{h}, \quad Y_{\delta}^{(\xi)}(t_j) = \frac{P_K \xi_{\delta}(t_j)}{h\sqrt{gh}}, \quad Y_{\delta}^{(G)}(t_j) = \frac{G_{\text{ref}}(\eta_{\delta}(t_j); h) \xi_{\delta}(t_j)}{\sqrt{gh}}, \quad (91)$$

where $(\eta_{\delta}, \xi_{\delta})$ is the trajectory computed with GL2 step δ . Let $\mathcal{A} = \{\eta, \xi, G\}$ label the three displayed fields. For a case specification θ , the validation discrepancy was

$$E(\theta; \delta) = \max_{\nu \in \mathcal{A}} \frac{\max_j \left\| Y_{\delta/2}^{(\nu)}(t_j) - Y_{\delta}^{(\nu)}(t_j) \right\|_{2,L}}{\max_j \left\| Y_{\delta/2}^{(\nu)}(t_j) \right\|_{2,L} + 10^{-12}}. \quad (92)$$

The maximum over time in the denominator prevents a temporary cancellation of one field from producing a large relative error. This is the historical pre-adjustment revision-2 panel, not the revision-3 category or cutoff comparison. Of the twelve cases, ten produced complete trajectories under both steps, and all ten satisfied

$$E(\theta; 0.01) \leq 1.126337 \times 10^{-5}.$$

The remaining steep Tanaka and equation-(33) Benjamin–Feir cases became incomplete at nearly the same physical times under both steps. Halving the step therefore neither changed an acceptance decision nor rescued either calculation, and no 0.0025 calculation was needed. Together with the May study, this panel validates the common step $\delta = 0.01$. It does not justify repeating the half-step calculation for every production case. On this panel the reported integration times were 4026.06 seconds for the 0.01 calculations and 7798.47 seconds for the 0.005 calculations. The pair therefore used 11824.53 seconds of integration work, whereas the 0.01 arm alone used 34.0% of that total. Repeating the validation pair in production would nearly triple the measured work of the selected method.

6.5 Historical JONSWAP/TMA spatial-cutoff calculation

Before revision 4 changed the spectral population, the random-sea cutoff was chosen on a predeclared shallow revision-3 sentinel. The two independently adjusted arms used 2048 points, order four, padding factor eight, step 0.01, saved spacing 0.08, residual tolerance 10^{-8} , and an iteration cap of five. Over the complete autonomous history, the largest relative whole-field differences between $K = 704$ and $K = 768$ were

$$(e_\eta, e_\xi, e_{G\xi}) = (4.64267, 1.93035, 7.91931) \times 10^{-4}. \quad (93)$$

The fields were transferred by the normalized coefficients in (13), and the common order-six target was recomputed on 1024 points. The internal order-four DNO output was not resampled. All three errors in (93) are below 10^{-3} , while the corresponding $K = 640$ versus $K = 768$ comparison exceeds 10^{-3} . Diagnostics restricted to modes $96 \leq |k| \leq 128$ are about 3×10^{-3} , so this calculation certifies only the global whole-field comparison, not a bandwise bound. Some stage residuals lie close to 10^{-8} , and this is a single worst-case sentinel. It supports the inherited $K = 704$ evolution method but does not validate the revision-4 relative-frequency population. It therefore did not replace the current revision-4 gate in Section 5.

The then-current revision-3 source was subsequently replayed from the archived full internal $K = 704$ endpoint. It reproduced all 289 autonomous saved frames with maximum relative L^2 errors $(3.99 \times 10^{-16}, 4.66 \times 10^{-16}, 1.17 \times 10^{-15})$ in $(\eta, \xi, G(\eta)\xi)$. The replay was accepted, solved every stage, had maximum residual 9.994198×10^{-9} , Hamiltonian drift 1.831187×10^{-5} , minimum water column 0.0387699, and CPU wall time 197.355 seconds. This verifies that historical source and the inherited evolution method against the archived sentinel; it does not validate the revision-4 relative-frequency population.

6.6 Whole-case production acceptance

Production consequently performs one rollout per attempted trajectory case, with $\delta = 0.01$, and stores candidate states every $\Delta s = 0.08$. After the initial condition has passed its declared family support, there is one whole-trajectory acceptance condition:

The fixed numerical method must return a complete admissible trajectory on the requested interval $[0, T]$.

Here T is the horizon declared by the case. “Complete admissible trajectory” means that every scheduled GL2 step reaches T , every implicit stage satisfies $R_n \leq \tau_{\text{GL}} = 10^{-8}$, every saved state is finite and satisfies $h + \eta > 0$ on the numerical grid, and every delivered target $G_{\text{ref}}(\eta; h)\xi$ at a saved time is finite. Failure to reach T , a nonfinite field or target, an unsolved stage, and loss of the water-wave graph domain are diagnostic causes of this one failure condition; they are not separate

fitted filters. If the condition fails, the complete case is rejected and no finite prefix enters the corpus. There is no per-case 0.005 comparison, discrepancy threshold, or 0.0025 retry.

Benjamin–Feir revision-4 and JONSWAP/TMA revision-4 cases also require saved-frame health in their full autonomous internal band. At every autonomous saved time t_j , before projecting for delivery, let $(N_F, K_F) = (1024, 256)$ for Benjamin–Feir and $(N_F, K_F) = (2048, 704)$ for JONSWAP/TMA, set $\Delta x_F = 2\pi/N_F$, and define

$$\mathcal{H}_F(t_j) = \frac{\Delta x_F}{2} \sum_{\ell=0}^{N_F-1} \left[\xi_\ell(t_j) (G_4^{K_F}(\eta(t_j); h) \xi(t_j))_\ell + g \eta_\ell(t_j)^2 \right], \quad (94)$$

where $G_4^{K_F}$ is the order-four DNO evaluated from the complete sharp internal state. Let ε_{64} be the smallest positive normal float64 number and set

$$D_{\mathcal{H},F} = \max_j \frac{|\mathcal{H}_F(t_j) - \mathcal{H}_F(0)|}{\max\{|\mathcal{H}_F(0)|, \varepsilon_{64}\}}. \quad (95)$$

These two families require $D_{\mathcal{H},F} \leq 10^{-3}$. At the same dense saved frames, the executor requires the complete internal state and $G_4^{K_F}(\eta; h)\xi$ to be finite and the minimum water column $\min_\ell\{h + \eta_\ell\}$ to be positive. It persists the initial Hamiltonian, maximum relative drift, minimum water column, and the state- and DNO-finiteness indicators. For JONSWAP/TMA, $t = 0$ and the Hamiltonian reference in these formulas are the full-band adjustment endpoint. No Hamiltonian gate is applied during the nonautonomous adjustment.

There is no production gate based on sign changes, roughness, slope, extrema, high-band energy, absolute target size, or visual appearance. Hamiltonian drift enters only through (95) for the two stated autonomous flows.

For a batch of B initial conditions, the production executor advances all B cases with $\delta = 0.01$. The eight GL2 steps between consecutive saved times are carried out inside this one batched calculation. Random-sea cases are grouped by their adjustment and autonomous horizons. Each case and its stage record are restricted to that case’s declared prefix before completeness and admissibility are decided. Values beyond a shorter case’s horizon do not belong to that case.

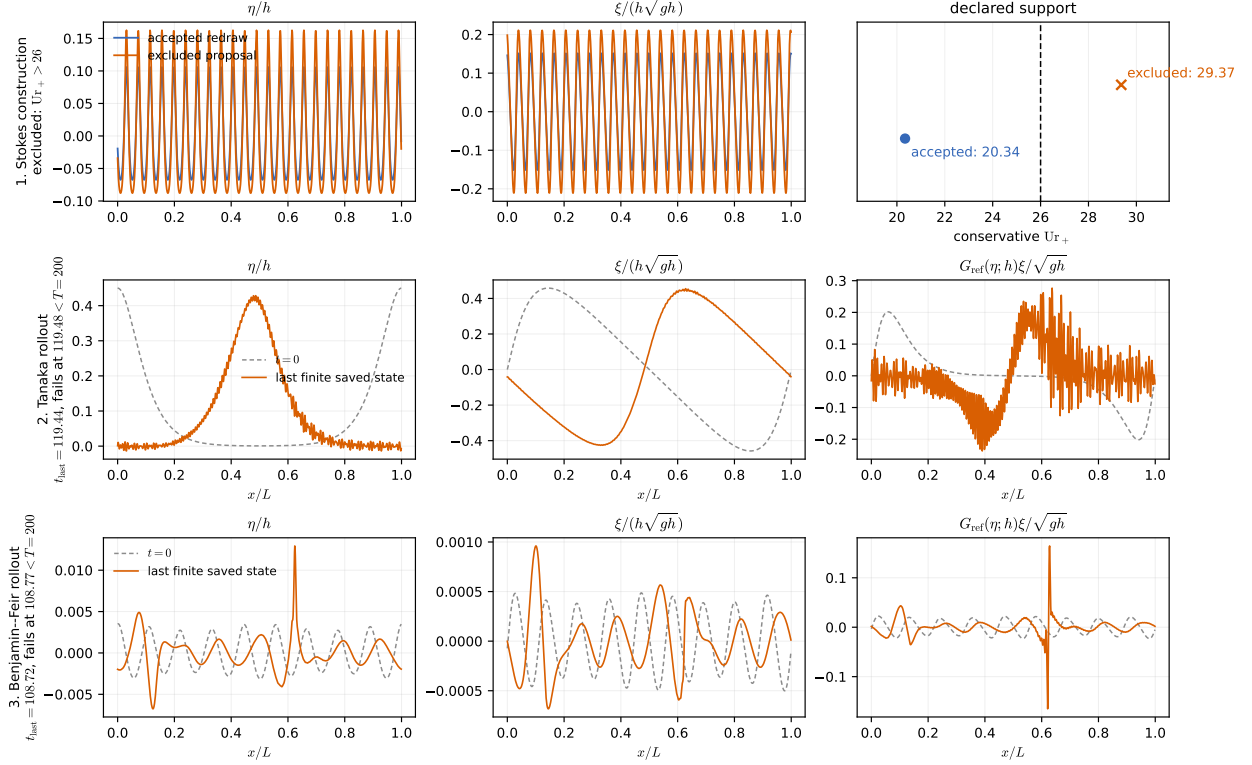


Figure 4: Historical illustrations of the two decision types retained in the current contract. In the top row, a revision-1 finite-depth Stokes support redraw first proposes an amplitude with $Ur_+ = 29.37 > 26$. That amplitude is outside the declared Stokes support. With the carrier, depth, phase, and steepness category unchanged, the next amplitude has $Ur_+ = 20.34$ and is accepted. Both displayed profiles are finite; the decision is the stated support inequality, not their appearance. The middle and bottom rows use the $\delta = 0.01$ arms of two July method-panel trajectories, both requested through $T = 200$; they predate the Tanaka revision-3 and Benjamin-Feir revision-4 executors and are not current production attempts. The Tanaka trajectory has its last finite saved state at $t = 119.44$ and first failed GL2 stage at $t = 119.48$. The Benjamin-Feir trajectory has its last finite saved state at $t = 108.72$; its stage at $t = 108.77$ reaches eight iterations with residual $3.8931 \times 10^{-7} > 10^{-8}$. These profiles illustrate the semantics of whole-trajectory failure only; they do not establish the rejection of a current corpus case or estimate a production rejection probability. No half-step calculation enters this figure or either decision.

Static Stokes states are not trajectories and do not use the complete-trajectory condition. A static case is retained only when its declared support produces a finite state with positive water column and a finite common target. The three-grid, cross-family spatial audit in the preceding subsection instead validates the constructor and common production grid before generation; it is not a case-level filter.

The implicit stage tolerance $\tau_{\text{GL}} = 10^{-8}$ remains part of the definition of the GL2 algorithm in the preceding subsection. It specifies when a stage equation has been solved, just as a linear-solver tolerance does; it is not a second data-quality classifier. Likewise, the only water-column condition used here is the mathematical graph requirement $h + \eta > 0$. The earlier half-depth margin $h + \eta \geq h/2$ is not part of the contract.

6.7 Additional historical and method-panel diagnostics

For grid index ℓ , write $\eta_\ell = \eta(x_\ell)$ and $\xi_\ell = \xi(x_\ell)$. Earlier solver studies and the fixed resolution panels monitored the following discrete energy in float64, using the unprojected order- M DNO output evaluated from the internal filtered state:

$$\mathcal{H}_M^N(z) = \frac{\Delta x}{2} \sum_{\ell=0}^{N-1} \left[\xi_\ell (\mathcal{G}_M^{N,p}(\eta; h) \xi)_\ell + g \eta_\ell^2 \right], \quad (96)$$

not reconstructed from the stored float32 $G_{\text{ref}}(\eta; h) \xi = P_K^\circ [\mathcal{G}_M^{N,p}(P_K \eta; h)(P_K \xi)]$ target. Because the implemented vector field includes spectral filtering and a truncated DNO, $\mathcal{H}_M^N$ is not assumed to be an exact invariant of that discrete evolution; its drift is a numerical diagnostic. Let z_n be the state after time step n , and set the fixed denominator floor $\varepsilon_{\mathcal{H}} = 10^{-14}$. The reported statistic is

$$D_{\mathcal{H}} = \max_n \frac{|\mathcal{H}_M^N(z_n) - \mathcal{H}_M^N(z_0)|}{|\mathcal{H}_M^N(z_0)| + \varepsilon_{\mathcal{H}} g h^2 L}. \quad (97)$$

This dimensionally floored statistic is reported on historical numerical panels together with mass, spectra, and stage iteration counts. It is not the current production definition: autonomous Benjamin–Feir and JONSWAP/TMA use the full internal-band quantity in (95). Mass, spectral occupancy, sign counts, and stage iteration counts remain diagnostics only. Internal Hamiltonian, state, DNO, and water-column telemetry are required production fields only for those two current families.

6.8 Storage after acceptance

Every requested storage time is a GL2 step endpoint: consecutive saved times are separated by eight steps of length 0.01. No temporal interpolation is used. Stored frames are selected only after the numerical method has returned a complete admissible trajectory, so a failed trajectory never contributes isolated training rows. The production step and saved spacing are fixed by (90); they are not adjusted case by case.

7 Reference-method and surrogate validation

Xu and Guyenne [4] organize validation around direct DNO tests, truncation order, dealiasing, long-time traveling waves, invariants, and time reversal. The replacement generator adopts the same separation of questions.

For two states $z = (\eta, \xi, h)$ and $\tilde{z} = (\tilde{\eta}, \tilde{\xi}, h)$ at the same depth, define the dimensionless state norm

$$\|(\eta, \xi)\|_h^2 = \|\eta/h\|_{2,L}^2 + \left\| \xi/(h\sqrt{gh}) \right\|_{2,L}^2, \quad (98)$$

where the normalized spatial norm $\|\cdot\|_{2,L}$ is defined in (3). Define their relative state error by

$$\text{rel}_h(z, \tilde{z}) = \frac{\|(\eta - \tilde{\eta}, \xi - \tilde{\xi})\|_h}{\|(\tilde{\eta}, \tilde{\xi})\|_h + 10^{-12}}. \quad (99)$$

For a validation step $\delta > 0$, let z_δ and $z_{\delta/2}$ denote trajectories sampled at common comparison times t_ℓ , where ℓ indexes those times. Define the method-level temporal defect by

$$E_t(\delta) = \max_\ell \text{rel}_h(z_\delta(t_\ell), z_{\delta/2}(t_\ell)). \quad (100)$$

For a traveling profile, let $\sigma \in [0, L]$ denote a horizontal translation and define

$$E_{\text{shape}}(t) = \min_{\sigma \in [0, L]} \frac{\|\eta(\cdot, t) - \eta(\cdot - \sigma, 0)\|_{2,L}}{\|\eta(\cdot, 0)\|_{2,L} + 10^{-12}}, \quad (101)$$

and obtain the numerical displacement from the minimizing σ . Starting from z_0 , integrate to time T and call the result z_f . Starting from z_f , use the same step lengths in reverse order with negative sign and call the returned state z_b . The forward-backward defect is

$$E_{\text{rev}}(T) = \text{rel}_h(z_b, z_0). \quad (102)$$

Thus the forward and backward calculations use exactly the same step lengths.

Every appearance of $D_{\mathcal{H}}$ in Table 3 refers to the historical or fixed method-level diagnostic defined in (97); it is not a row, case, or trajectory acceptance condition.

Successive- M differences on general states are called order sensitivity, not convergence to the physical DNO, unless an independent boundary-integral or conformal reference is supplied. This qualification matters because Xu-Guyenne report an amplitude- and resolution-dependent optimal order: larger M need not reduce error once the recursion is ill-conditioned.

The completed three-grid Tanaka study remains useful evidence that periodic wrapping removes the old clipped-profile defect. It does not by itself validate the four training families, the three diagnostic suites, the time integrator, or the choice $M = 6$.

8 Temporal sampling, splitting, and normalization

Temporal selection is performed only after the fixed-step calculation has returned a complete admissible trajectory. Tanaka is evaluated on the dense time grid

$$t_j = 0.08j, \quad j = 0, \dots, 2500, \quad (103)$$

and 200 times are retained for each case. The internal GL2 step is 0.01; 0.08 is the spacing of the candidate storage grid.

For Tanaka, define the discrete surface-gradient energy and its activity by

$$S_{T,j} = \sum_{\ell=0}^{N-1} |\partial_x \eta(x_\ell, t_j)|^2, \quad A_{T,j} = \frac{|\delta S_{T,j}|}{S_{T,j} + 10^{-12}}. \quad (104)$$

Table 3: Validation objects, measured quantities, and decision roles.

Panel	Quantity reported	Role
Manufactured DNO	Exact-DNO error, discrete-target error, learned-target error, and end-to-end error as defined in Appendix B.2.	Selects the discrete reference configuration, not network weights.
Three-grid construction	$\Delta_{\text{IC},K}$ and $\Delta_{G,K}$ from (84)–(86).	Required before production; validates each constructor and the common spatial configuration.
Time-step study	$E_t(\delta)$, maximum stage residual, and $D_{\mathcal{H}}$.	Required before production; selects one common GL2 step. It is a method-level panel, not a calculation repeated for each corpus case.
Traveling Stokes/Tanaka	$E_{\text{shape}}(t)$, displacement divided by t , mass, (97), and $E_{\text{rev}}(T)$.	Validates long-time propagation of the reference method.
Benjamin–Feir	Modal amplitudes $2 \hat{\eta}_n(t) $ at the carrier and sidebands, recurrence error, $D_{\mathcal{H}}$, and high-band energy.	Validates modulation dynamics; growth alone is not labeled breaking.
Airy	Flat-surface identity and cubic small-amplitude remainder scaling.	Post-freeze architecture verification only.
Sparse Fourier	One-step DNO error and complete rollout error; nonfinite rollouts count as infinite error.	Development cases select only the corpus; final cases only report.
Support sensitivity	Cellwise attempted/accepted counts under predeclared neighboring family-support bounds.	Tests whether a declared support bound removes a difficult parameter category; it does not alter the fixed production step.

Here

$$\delta S_0 = S_1 - S_0, \quad \delta S_j = \frac{S_{j+1} - S_{j-1}}{2} \quad (1 \leq j \leq 2499), \quad \delta S_{2500} = S_{2500} - S_{2499}.$$

The derivative $\partial_x \eta$ is computed by the Fourier multiplier ik_n on the common grid. Thus Tanaka receives extra samples when the spatial surface derivative changes rapidly during crest interaction.

Convolve the nonnegative Tanaka activity values with a normalized discrete Gaussian, extending the endpoint values outside the time interval. The Gaussian standard deviation is 50 candidate-time steps, and its support is truncated at three standard deviations, rounded upward to an integer number of time indices. Call the smoothed values $\tilde{A}_{T,j}$ and define

$$p_{T,j} = \frac{1 - \alpha_T}{2501} + \alpha_T \frac{\tilde{A}_{T,j}}{\sum_{r=0}^{2500} \tilde{A}_{T,r}}, \quad \alpha_T = 0.5. \quad (105)$$

If the smoothed activity vanishes identically, its normalized term is replaced by the uniform distribution. The 200 retained indices are the deterministic inverse-CDF locations of the midpoint quantiles $(\ell + 1/2)/200$, $\ell = 0, \dots, 199$. The first and last candidate times are included, repeated

indices are advanced to the next unused index, and all selected indices and activity parameters are stored with the case. This rule depends only on the accepted reference trajectory, never on model error.

For a Benjamin–Feir trajectory, let $J_{\text{BF}} = T_{\text{BF}}/0.08$, where T_{BF} is defined in (50). Retain the 200 saved-grid indices

$$j_\ell = \left\lfloor \frac{\ell J_{\text{BF}}}{199} + \frac{1}{2} \right\rfloor, \quad \ell = 0, \dots, 199. \quad (106)$$

These indices are approximately uniform in time and include both endpoints. The Benjamin–Feir storage rule therefore introduces no additional event score.

For a random-sea trajectory, let

$$T_{\text{hor}} = 16 \frac{2\pi}{\omega_p}, \quad T_s = 0.08 \left\lfloor \frac{T_{\text{hor}}}{0.08} \right\rfloor, \quad J_s = \frac{T_s}{0.08}.$$

It retains the saved-grid indices

$$j_\ell = \left\lfloor \frac{\ell J_s}{15} + \frac{1}{2} \right\rfloor, \quad t_\ell = 0.08 j_\ell, \quad \ell = 0, \dots, 15.$$

Thus the integration ends at the last saved-grid point not exceeding sixteen peak periods, and the retained endpoints are 0 and T_s . For JONSWAP/TMA, this $t = 0$ is the complete 2048-point, $K = 704$ adjustment endpoint and the start of the autonomous clock, not the initial linear random sea. Static Stokes cases retain only $t = 0$, with phase already contained in the case record.

Random-sea cases in one numerical batch can have different values of T_s . The single production batch is advanced to the longest required saved-grid endpoint in the batch, but each case’s fields and GL2 stage record are cropped to its own declared prefix before completeness and admissibility are decided. Hence values after a shorter case’s horizon cannot affect its acceptance. A regression test obtains the same decision, retained fields, and prefix stage record in serial and batched runs even when post-horizon values are deliberately made nonfinite. Batch size is recorded per launch. The completed Tanaka revision-3 gate used batch sizes six and five for its complementary chunks. The two excluded historical 1024-case training shards used batch size 256; this changed transaction granularity, not the population law.

Training, validation, and test cases come from three disjoint tagged PCG64 construction streams. Their root seeds are respectively 2026072210, 2026072204, and 2026072205. Every attempted case stores its root seed, stream identifier, and realized parameters. Let s be the split root, let f identify the physical family, let v identify the generator revision, let r be the stream identifier, and let a be the attempted-case index. Every family initializes its per-case PCG64 stream from the five integer words (s, f, v, r, a) . Thus replay of one attempted case reproduces its complete specification without depending on the number of earlier accepted cases. Split membership is therefore fixed when the attempted case specification is written, before integration or time selection; it is not assigned from the resulting trajectory. The training stream continues until it has the post-acceptance count C per family selected by the learning curve. The validation and test streams each contain 1,024 accepted cases per family, balanced over the same parameter categories to within one case. Every rejected attempt remains attached to its original split manifest. A nested training chunk is never enlarged in place. The successive (C_0, A) pairs, where C_0 is the preceding cumulative count and A is the new count per family, are

$$(0, 2048), (2048, 2048), (4096, 4096), (8192, 8192).$$

They use distinct stream identifiers 0, 1, 2, 3 and the category increments in (67). Validation and test each use one fixed, non-nested chunk. A requested count in one family-split category is never filled by copying or silently drawing from another category. In the static Stokes generator, one candidate fixes the mode, depth, phase, and steepness category and permits at most 1000 amplitude redraws within that same category. The sampler record contains every rejected amplitude and its Ur_+ value, followed by the accepted amplitude on success. If all 1000 redraws are exhausted, the sampler raises with the complete strict record. The implemented static executor stores that record as a zero-row out-of-support attempt and schedules a replacement in the same category. It is not a fatal batch error. An unclassified error propagates without being relabeled; only the trajectory executor’s explicit fatal exception writes a terminal failure record. This internal amplitude-redraw limit does not consume additional outer attempts. The replacement case does consume the next durable attempt under (69).

No two temporal snapshots from the same case can appear in different subsets. This removes exact temporal-sibling leakage; it does not imply that nearby parameter draws are statistically independent.

The loader implements the case-balanced rule stated in Section 1: it selects exactly one stored time from every accepted training case in every epoch and batches those selected rows. The validation time for each case is fixed deterministically. A different number of stored times requires a separate learning-curve comparison; it is not chosen from rollout failures.

Let $\mathcal{I}_{\text{train}}$ denote all stored rows from accepted training cases. For row $i \in \mathcal{I}_{\text{train}}$, let η_i and ξ_i be its two input fields, let h_i be its depth, and let $G_i^* = G_{\text{ref}}(\eta_i; h_i)\xi_i$ be its target from (11). The network retains the scale normalization used by the strongest current checkpoint, but estimates it on this set only:

$$s_\eta = \max_{i \in \mathcal{I}_{\text{train}}, x} |\eta_i(x)|, \quad s_\xi = \max_{i \in \mathcal{I}_{\text{train}}, x} |\xi_i(x)|, \quad s_G = \max_{i \in \mathcal{I}_{\text{train}}, x} |G_i^*(x)|. \quad (107)$$

Let $h_\star = L/(2\pi)$ be the fixed reference length. For one row, write $(\eta, \xi, h, G^\star)$ for its two input fields, depth, and target. The network receives

$$\tilde{\eta} = \eta/s_\eta, \quad \tilde{\xi} = \xi/s_\xi, \quad d_h = \log\left(\frac{h}{h_\star}\right), \quad \tilde{G}^\star = G^\star/s_G. \quad (108)$$

Hence d_h is dimensionless and is unchanged by the period rescaling in Appendix C; on the production domain $L = 2\pi$ it has the same numerical value as the existing input $\log h$. The authenticated training-only values are $s_\eta = 0.15583430230617523$, $s_\xi = 0.09394174814224243$, and $s_G = 0.12198150902986526$, computed from 6,832,128 rows. The release audit verifies all three before training. The current loader nevertheless uses one as a defensive fallback if a nonpositive scale reaches it; the writeup does not count that fallback as validation of a malformed corpus. Their statistics fingerprint is `4e8ec96947bc05cee3b375467499b0ae00a7814f9982614084efd859efe7c3b8`. No per-family, per-mode, validation, or diagnostic statistic enters (107). The cached scales are bound to the checked identity of the assembled dataset. Reports include both the number of case units and the number of stored snapshots. Corpus size is selected by a learning curve in accepted case count, rather than by reproducing the v9 snapshot count by default. The Airy, manufactured, and sparse diagnostic cases are absent from all three training subsets and from the training-scale calculation. If a diagnostic result motivates a redesign, the final paper reports a fresh, case-disjoint diagnostic draw after the revised model is frozen.

At C accepted training cases per family, the case-balanced loader presents $4C$ samples per epoch. With global batch size 1024, comparisons at a fixed epoch count would therefore give larger corpora more optimizer updates. The learning curve instead uses 10,240 optimizer updates:

C	Steps per epoch	Epochs for 10,240 updates
2,048	8	1,280
4,096	16	640
8,192	32	320
16,384	64	160

The learning-rate schedule is a function of optimizer step, not epoch. Checkpoints are compared at the same predeclared update numbers; the number of epoch boundaries is not allowed to give the smaller corpus more checkpoint-selection opportunities. Test errors are first aggregated by case within each family and then with equal weight across the four families. A single statistic weighted by the 1, 200, 200, and 16 stored rows per case is not a paper-level four-family result.

9 Execution plan

The target, four population laws, spatial window, reference integrator, complete-trajectory condition, storage rule, and bounded quota mechanism are frozen. Before the production executor was simplified to one time step, an exact-contract four-family smoke used run-specification schema 1 and computed the 0.01/0.005 validation pair:

Family	Accepted	Rows	Pair discrepancy	Wall time (s)
Stokes	1	1	—	3.782
Tanaka	1	200	3.1953×10^{-8}	5,555.35
Benjamin–Feir	1	200	3.7058×10^{-8}	5,561.11
JONSWAP/TMA	1	16	7.6600×10^{-7}	835.19

The four cases assembled into one checked dataset view containing 417 finite rows on 1024 points, and the loader selected one time from each case. The pair discrepancies are historical method-validation evidence, and the recorded wall times include work that production no longer performs. These run-spec-schema-1 artifacts predate both the bounded scheduler and Tanaka revision 3. They deliberately cannot resume as a current production run.

Generator revision is a family-specific identity, not one global integer. The present revision map is

$$(v_{\text{Stokes}}, v_{\text{Tanaka}}, v_{\text{BF}}, v_{\text{sea}}) = (2, 3, 4, 4). \quad (109)$$

Here v_{BF} is the Benjamin–Feir revision and v_{sea} is the JONSWAP/TMA revision. Revision-1 smokes and all superseded revision-2 trajectory artifacts remain historical constructor, transaction, or method-selection evidence. They cannot be resumed into an output directory with a different family revision or counted toward its quota. A combined dataset view requires exactly one revision per family. Within each (family, revision) pair it requires one numerical execution identity and one generator-source identity; it does not require the filtered Tanaka order-six execution record to equal the sharp Benjamin–Feir or JONSWAP/TMA order-four records.

Current revision-2 Stokes evidence includes one exact case from each of the four Stokes categories. All four were accepted, their quality-decision masks were zero, and the complete calculation took 4.6875 seconds. Its production splits are also complete at 16384 training, 1024 validation, and 1024 test cases; they are preserved. The older reduced real-GL2 transactions from the trajectory families check historical software wiring only. For Tanaka revision 3, the full-horizon boundary calculation in Section 6 has frozen the amplitude-first $R \geq 10$ support and $K_{\text{evol}} = 256$ Hou–Li execution contract. Before this Tanaka category pilot, 50 focused contract/manifest tests, 212 generator tests, and 102 script tests passed.

At 16:09 on 31 July 2026, the exact Tanaka revision-3 production pilot was launched in two complementary chunks. The first requests category codes 0–5 and the second requests codes 6–10, so together they request one case from every one of the eleven Tanaka categories. Their durable batch-zero proposals were written before construction. Both chunks use (36), $K_{\text{evol}} = 256$, (73), delivered P_{128} rows and labels, GL2 step 0.01, and only the construction-domain and complete-trajectory decisions stated above. Both chunks completed. The first recorded 6 accepted, 6 attempted, and 0 rejected cases; the second recorded 5 accepted, 5 attempted, and 0 rejected cases. Together they cover cell codes 0–10 exactly once and contain eleven distinct case identities. Every trajectory reaches $T = 200$, has 2501 saved frames, retains 200 selected rows, keeps a positive water column, and solves every GL2 step. The maximum relative stage residual over all eleven cases is 9.997×10^{-9} . Every case has required, evaluated, and failed quality masks respectively equal to 768, 791, and 0.

The configuration fingerprints of the two completed pilot chunks are, respectively,

```
3a107cc1e4a711f48c35a0f1995f484552b658f0f926ee324e487cee940fbda8
c089df6be04f25c4c59794f0e34599ee457e0db076e807834608fed1c5c62b9f.
```

Their summary-file SHA-256 digests are, in the same order,

```
075dd7c30d325fc5034b6ee9350949a99317ae22e1656c4597f93d4de4cfe164
cd24c7a801150667b5872332baa38353865f4ae04349c9cfd75ae4f1fc87e7df.
```

The two summaries carry identical numerical, dependency, and source-hash tables, and those hashes match the frozen files. The measured wall times were 1493.31 seconds for the six-case chunk and 2931.72 seconds for the five-case chunk. Their 2200 validation-tagged pilot rows remain outside the final corpus.

This result freezes Tanaka revision 3 as generation-ready with the conditional $R \geq 10$ support, one sharp P_{128} initial projection, a zero-filled $K_{\text{evol}} = 256$ workspace, the power-36 Hou-Li multiplier after every completed step, and delivered P_{128} rows and labels. It adds no Hamiltonian, sign-count, slope, roughness, or spectral rejection rule.

At 17:02 on 31 July 2026, two complementary 1024-case Tanaka training chunks were launched for cumulative accepted intervals 0:1024 and 1024:2048, with separate streams, batch size 256, and configuration fingerprints

```
d4b38dea49ade42f0251aeb993c935155cd07a516475a2d922c48226ca4973c1
ccd5968fcb936501245e5b1858282b953fdfee9e65d027e66bc323783d38da3f.
```

They use the frozen revision-3 contract above. Both training chunks and the corresponding 1024-case validation and test chunks completed, with all 4096 attempts accepted and 200 rows retained per case. These artifacts are now historical and are excluded from the paper corpus.

The old solitary-wave solve had an effective component-amplitude floor of 9.9478×10^{-4} . Exactly 64 of the 4096 cases request one component below this floor; the worst is inflated by $41.27\times$ after the complete periodic construction and projection. The old trajectory rows are therefore excluded. The corrected population remains revision 3 but draws a fresh deterministic population with 16384 training cases and 1024 cases in each of validation and test; it does not reuse the historical random specifications.

The source-authenticated release is complete. Stokes retained 18432 cases from 18432 attempts and stored 18432 rows; corrected Tanaka retained 18432/18432 and stored 3686400 rows; Benjamin-Feir retained 18432/18455 and stored 3686400 rows; and JONSWAP/TMA retained 18432/18726 and stored 294912 rows. The aggregate is 73728 accepted cases from 74045 attempts, 317 zero-row rejections, and 7686144 rows. The exact cumulative combined-summary SHA-256 identities at 2048, 4096, 8192, and 16384 training cases per family are 2265cb71270accb3f6d9103e90433b664ca813989dfab63403d8612af22d28c3, 7feb018ee4d94eab10278917c145262026cccab6169eaaa2

d8877f7c7b28c9da, 43b1eb91198e724c6f32b871c2d0efd0d8355533789919df5897ad588e7e85f6, and 32f764cf865c90892ee0329e48fe992cbc24df2707e05fd1af2ee9cd2b84307c. The immutable document handoff reports the exact attempted, accepted, and rejected marginal for every declared family, split, and parameter category, thereby closing the reporting contract above. Its SHA-256 identity is

b21830cf56f9c17a1ee165624c71fdc0e286727747d364330fbace19e5ef6053.

No reported model has been trained on this corpus. The remaining empirical work is:

1. **Run the complete nested learning curve.** Extend the training split additively to

$$C \in \{2048, 4096, 8192, 16384\}$$

accepted cases per family while keeping validation and test fixed. The final point contains 7,686,144 retained rows, comparable to historical v9’s 7,633,989 snapshots. Train two seeds at every C with the same architecture, optimizer, and 10,240-update schedule. Compare final checkpoints at that common update count. Report every point rather than stopping at an undefined “material improvement” threshold.

2. **Test shallow mixed-mode coverage.** Evaluate every learning-curve endpoint on the 11 historical shallow-steep cases and the frozen sparse development panel. The existing checkpoint C27, trained on all v9 sources, is an external performance baseline, not a matched ablation. Measure relative terminal elevation error and assign infinite error to every nonfinite rollout. Report the median, 95th percentile, nonfinite count, and counts above 0.25, 0.50, 0.75, and 1. A failure motivates a separately versioned population revision; it does not change this frozen corpus after results are seen.
3. **Freeze and test once.** After all models and checkpoints are fixed, evaluate the untouched training-family test cases, Airy panel, manufactured-DNO panel, and final sparse panel. Report errors by case within family and then weight the four family summaries equally. Any later design change requires a new final sparse draw.

10 Claims this plan does and does not support

The completed corpus supports the following statements:

- every attempted initial condition is reproducible from a complete, finite-dimensional case specification and one common configuration;
- every training label uses the same discrete operator convention;
- training, validation, and test rows come from disjoint cases;
- no Airy, manufactured-harmonic, or sparse-Fourier diagnostic row contributes to optimization or empirical normalization;
- every saved state of an accepted reference trajectory remains in the water-wave graph domain, and the trajectory is complete under the declared fixed GL2 method; and
- the declared four-family depth, nonlinearity, and spectral coverage is reported in physical dimensionless variables.

It does not, without further analysis, support the stronger statements that the selected G_M equals the physical DNO, that Hamiltonian conservation proves trajectory accuracy, that JONSWAP spans every physically relevant sea, or that a trajectory rejected by this numerical method is physically impossible. Success on the four training families also does not establish arbitrary mixed-mode generalization or accuracy below the declared shallow-TMA depth; those remain empirical questions measured by the diagnostic suites. The existing C27 result cannot answer the corpus-ablation question because C27 was trained on all v9 sources. The order, grid, timestep, benchmark, and new-corpus retraining studies are included precisely to keep those questions separate.

A Audit of the current v9 archive

The archive `combined_dataset_v9.npz` contains 7,633,989 snapshots on $N = 1024$ points over $L = 2\pi$. Its six construction categories are Gaussian seas (460,740 rows), manufactured single modes (500,000), fifth-order Stokes states (1,000,000), Tanaka-profile states (2,200,000), Benjamin–Feir states (2,033,249), and sparse Fourier packets (1,440,000). The suffixes `g0`, `g1`, and `modal` name generator runs, not different initial-condition families or DNO orders.

The archive remains a valuable empirical baseline. It was assembled from generators with different row counts, filtering rules, and target conventions. The relevant distinctions are summarized in Table 4.

Table 4: Why the flat v9 archive does not define the replacement paper corpus.

Object	Current archive	Replacement rule
Sampling rule	Source-dependent numbers of stored times make snapshot weight differ from case weight.	In every epoch select one stored time from every accepted case, then batch those selected rows.
Discrete target	Static sources retain the direct order-six output, whereas rollout sources are projected to $ k \leq 128$.	Use (11) for every row.
Acceptance	Early sources were filtered rowwise; later sources used source-specific whole-case Hamiltonian tests.	Use declared support for static Stokes states and one complete fixed-step numerical-path condition for each trajectory family. Benjamin–Feir revision 4 and autonomous JONSWAP/TMA revision 4 additionally require the declared internal Hamiltonian bound (95).
Reconstructibility	Some phase arrays and case fields were not stored, and no stage residual was recorded.	Store every parameter and test the residual (83).
Statistical split	A rowwise split can place temporal siblings in different subsets; normalization used the complete archive.	Split by attempted case and fit scales on accepted training cases.
Support changes	Some failed parameter combinations were replaced by hand, for example one Benjamin–Feir mode pair.	Declare support first; record rejection without changing parameters.

Table 5 records the historical parameter envelopes needed for the direct comparison. These are support bounds, not empirical quantiles, and independent dimensional draws did not induce uniform density in the dimensionless coordinates used below. Here H_s is significant wave height, k

is wavenumber, k_p is peak wavenumber, k_c is carrier wavenumber, b is spectral bandwidth, and n , n_c , and n_j are Fourier mode indices. The symbols a and a_j are wave amplitudes, Δn is sideband separation, and $\rho_{\pm}$ are sideband-to-carrier amplitude ratios.

Table 5: Historical support envelopes retained for comparison.

Construction	Declared support in v9
Gaussian sea	$H_s \in [0.005, 0.03]$, $k_p \in [2, 12]$, bandwidth $b \in [1.5, 5]$; $h \in [0.1, 1.5]$ or $h \in [5, 25]$.
Manufactured mode	$n \in \{1, \dots, 20\}$, $h \in [0.02, 1.5]$, $kh \in [0.02, 20]$, and $0.005 \leq ka \leq 0.15$, subject to $a \in [10^{-4}, 0.011494]$.
Stokes	Deep: $n \in \{1, \dots, 20\}$, $h \in [4, 50]$. Finite: $n \in \{14, \dots, 26\}$, $h \in [0.02, 1.5]$, $kh \in [0.5, 5]$. Both impose $ka \leq 0.15$.
Tanaka	Main: $h \in [0.01, 0.30]$, one to three crests, total $\sum_j a_j/h \in [0.10, 0.35]$. Steep: $h \in [0.20, 0.35]$, one crest, $a/h \in [0.25, 0.45]$.
Benjamin-Feir	$h \in [0.5, 4]$, $n_c \in \{4, \dots, 20\}$, $k_c a \in [0.05, 0.13]$, $\Delta n \in \{2, 3, 4\}$, and $\rho_{\pm} \in [0.05, 0.20]$.
Sparse Fourier	One to five modes, $h \in [0.005, 0.5]$, $kh \in [0.2, 1.5]$, total $a/h \in [0.03, 0.25]$, and $\sum_j n_j a_j \leq 0.15$.

Two historical formulas matter for interpreting the redesign. First, the old sea uses the following Gaussian elevation spectrum rather than the JONSWAP/TMA construction in Section 4.4:

$$S_{\text{old}}(k; k_p, b) = \exp \left[-\frac{(|k| - k_p)^2}{2(k_p/b)^2} \right]. \quad (110)$$

Second, the source archived as *linear* is a manufactured harmonic trace. For wavenumber k , amplitude a , phase ϕ , and direction sign $s \in \{-1, 1\}$, set $\vartheta = kx + \phi$ and $\omega^2 = gk \tanh(kh)$. Then

$$\eta(x) = a \cos \vartheta, \quad (111)$$

$$\xi(x) = s \frac{ag \cosh(k(h + \eta(x)))}{\omega \cosh(kh)} \sin \vartheta. \quad (112)$$

The nonlinear dependence on η makes this an exact manufactured boundary trace, not a separate physical linear-wave family.

The historical depth union is approximately $0.005 \leq h \leq 50$. This is not automatically the support of the replacement dataset. The four-family design reaches $h \simeq 0.0083$ through its shallow random-sea stratum and $h = 50$ through Stokes; the smaller-depth interval remains diagnostic unless a separate resolution study admits it. The full source-by-source audit is therefore evidence for the redesign, not the main definition of it.

B Diagnostic specifications and separation from training

The three diagnostic suites are Airy, exact manufactured, and sparse Fourier cases. They have complete case specifications, but they are not used for training. The sparse-Fourier cases are split into disjoint development and final-test sets. Sparse development cases may reject or revise the replacement corpus through the predeclared gates in Section 9; they never contribute loss terms or normalization statistics. Final-test cases are generated from new case keys and remain untouched until the corpus, network, optimizer, and checkpoint are frozen. If a final-test result causes another design change, it ceases to be final and another case-disjoint test draw is required.

The Airy grid is a fixed analytic verification panel, not a random development/test draw, and is evaluated only after the network is frozen. The manufactured panel is separate: it may select the

reference discretization (M, N, p, K) because its purpose is to verify that numerical formula, but it never selects network parameters. No final diagnostic row contributes to optimization, empirical normalization, network-hyperparameter selection, or checkpoint selection.

B.1 Airy linear-limit panel

For the exact flat-surface test, choose an integer mode $n \geq 1$, set $k = k_n$, and choose a depth $h > 0$, trace amplitude $b_A > 0$, and phase $\psi \in [0, 2\pi)$. The identity is

$$\eta = 0, \quad \xi = b_A \cos(kx + \psi), \quad G(0; h)\xi = k \tanh(kh)\xi. \quad (113)$$

For paired phase-space tests, choose an elevation amplitude $a > 0$, phase $\phi \in [0, 2\pi)$, and direction sign $s \in \{-1, 1\}$. With $G_0(k; h) = k \tanh(kh)$ and $\omega(k, h) = \sqrt{gG_0(k; h)}$, define

$$\eta_A(x) = a \cos(kx + \phi), \quad (114)$$

$$\xi_A(x) = s a \frac{\omega(k, h)}{G_0(k; h)} \sin(kx + \phi). \quad (115)$$

These are valid Zakharov initial data but are not asserted to be exact nonlinear traveling waves at appreciable ka . Let $\mathcal{N}_A$ be the tested integer modes, let $\mathcal{X}_A$ contain the tested values of the dimensionless depth $\chi = kh$, and let $\mathcal{E}_A$ contain the tested values of the steepness $\epsilon = ka$:

$$\begin{aligned} \mathcal{N}_A &= \{2, 4, 8, 16, 24\}, \\ \mathcal{X}_A &= \{0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5, 10, 20\}, \\ \mathcal{E}_A &= \{10^{-4}, 10^{-3}, 10^{-2}, 0.03, 0.075, 0.15\}. \end{aligned} \quad (116)$$

For every $n \in \mathcal{N}_A$, $\chi \in \mathcal{X}_A$, and $\epsilon \in \mathcal{E}_A$, set

$$k_n = \frac{2\pi n}{L}, \quad h = \frac{\chi}{k_n}, \quad a = \frac{\epsilon}{k_n}, \quad (117)$$

and retain only categories satisfying

$$0.005 \leq h \leq 50, \quad \frac{a}{h} = \frac{\epsilon}{\chi} \leq 0.16. \quad (118)$$

Use both $s = \pm 1$ and phases $\phi \in \{0, \sqrt{2}\}$ radians; the latter are not related by an integer-grid shift on the $N = 1024$ grid. The flat test uses the same (n, χ) categories, $\psi \in \{0, \sqrt{2}\}$, and $b_A \in \{10^{-3}, 10^{-2}, 10^{-1}\}$. These rules determine (n, h, a, ϕ, s) or (n, h, b_A, ψ) completely and exclude bottom-crossing surfaces.

The amplitude ladder tests the declared small-surface structure of the *discrete target*; no independent nonlinear physical-DNO solver is assumed here. Define

$$\begin{aligned} r_A^*(a) &= G_{\text{ref}}(\eta_A; h)\xi_A \\ &\quad - P_K^\circ[G_0(h)P_K\xi_A + G_1(P_K\eta_A; h)P_K\xi_A]. \end{aligned} \quad (119)$$

Both η_A and ξ_A are proportional to a . Therefore $G_0(h)\xi_A$ has degree one in a , $G_1(\eta_A; h)\xi_A$ has degree two, and $G_m(\eta_A; h)\xi_A$ has degree $m + 1$. After subtracting the first two terms, the remainder therefore begins at degree three:

$$\|r_A^*(a)\|_{2,L} = O(a^3) \quad (a \rightarrow 0) \quad (120)$$

at fixed (k, h, ϕ, s) and fixed (M, N, p, K) . When the projected G_2 coefficient is nonzero, the predicted leading slope is three; a vanishing coefficient is reported rather than fitted on roundoff. The diagnostic is the least-squares slope of $\log \|r_A^*(a)\|_{2,L}$ against $\log a$ over the three smallest usable amplitudes. The learned remainder is defined by replacing $G_{\text{ref}}(\eta_A; h)\xi_A$ with the network output. Thus the flat-surface identity is a physical analytic test, whereas the nonlinear amplitude ladder tests consistency with the declared polynomial target and architecture. Thus this held-out panel is more informative about the analytic $G_0 + G_1 + O(\eta^2)$ architecture than a large collection of similar single-mode training rows.

B.2 Manufactured exact-DNO panel

For an integer mode $n \geq 1$, set $k = k_n$. Choose a depth $h > 0$, amplitude $a > 0$, phase $\phi \in [0, 2\pi)$, and direction sign $s \in \{-1, 1\}$, and set $\omega = \omega(k, h) = \sqrt{gk \tanh(kh)}$. Define

$$\varphi(x, y) = s \frac{ag \cosh(k(y+h))}{\omega \cosh(kh)} \sin(kx + \phi), \quad \eta(x) = a \cos(kx + \phi). \quad (121)$$

The boundary trace is exactly (112), and the normal derivative is known without solving a boundary-value problem:

$$[G_{\text{exact}}(\eta; h)\xi](x) = (\varphi_y - \eta_x \varphi_x)_{y=\eta(x)}. \quad (122)$$

This is the one-dimensional counterpart of the direct DNO test in Xu–Guyenne equations (30)–(32) [4]. For the exact benchmark, the full trace $\xi = \varphi|_{y=\eta}$ is evaluated on a sufficiently resolved grid and the unprojected $\mathcal{G}_M^{N,p}(\eta; h)\xi$ is compared with $G_{\text{exact}}(\eta; h)\xi$. This is distinct from the training target (11): in general

$$G(\eta; h)(P_K \xi) \neq G(\eta; h)\xi. \quad (123)$$

The frozen manufactured grid uses the same $n \in \mathcal{N}_A$, $kh \in \mathcal{X}_A$, $ka \in \mathcal{E}_A$, feasibility condition (118), phases $\phi \in \{0, \sqrt{2}\}$, and signs $s = \pm 1$ as the Airy panel. The constructor (121) then determines the full boundary trace and exact normal derivative, so every benchmark case is reconstructible.

Once the reference configuration is frozen, let G_{net} denote the frozen learned surrogate. Report four different errors:

$$e_{\text{CS}} = \frac{\|\mathcal{G}_M^{N,p}(\eta; h)\xi - G_{\text{exact}}(\eta; h)\xi\|_2}{\|G_{\text{exact}}(\eta; h)\xi\|_2}, \quad (124)$$

$$e_{\text{target}} = \frac{\|G_{\text{ref}}(\eta; h)\xi - P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}{\|P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}, \quad (125)$$

$$e_{\text{learn}} = \frac{\|G_{\text{net}}(P_K \eta; h)(P_K \xi) - G_{\text{ref}}(\eta; h)\xi\|_2}{\|G_{\text{ref}}(\eta; h)\xi\|_2}, \quad (126)$$

$$e_{\text{total}, K} = \frac{\|G_{\text{net}}(P_K \eta; h)(P_K \xi) - P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}{\|P_K^\circ[G_{\text{exact}}(\eta; h)\xi]\|_2}. \quad (127)$$

Here e_{CS} measures the unprojected order- M reference, e_{target} is the total discrepancy of the declared discrete target and therefore combines Craig–Sulem, grid, input-projection, and output-projection effects; it is not an isolated projection error. e_{learn} isolates fitting of the declared discrete target, and $e_{\text{total}, K}$ is the end-to-end fixed-band error. This prevents reference truncation or projection error from being credited to or blamed on the network.

B.3 Sparse-Fourier shallow stress suite

These cases test shallow mixed-mode generalization and are never used for training. Choose a depth $h > 0$, a mode count $m \in \{2, 3, 4, 5\}$, and distinct positive Fourier indices $n_1, \dots, n_m$. For each component, choose an amplitude $a_j > 0$, phase $\phi_j \in [0, 2\pi)$, and direction $s_j \in \{-1, 1\}$. The initial state is

$$\eta_F(x) = \sum_{j=1}^m a_j \cos(k_{n_j}x + \phi_j), \quad (128)$$

$$\xi_F(x) = \sum_{j=1}^m s_j a_j \frac{\omega(k_{n_j}, h)}{G_0(k_{n_j}; h)} \sin(k_{n_j}x + \phi_j). \quad (129)$$

Here $G_0(k; h) = k \tanh(kh)$ and $\omega(k, h) = \sqrt{gG_0(k; h)}$, as in the Airy construction. Fresh cases satisfy

$$0.02 \leq h \leq 0.12, \quad 0.2 \leq k_{n_j}h \leq 1.5, \quad 0.03 \leq \sum_{j=1}^m \frac{a_j}{h} \leq 0.18, \quad \sum_{j=1}^m k_{n_j}a_j \leq 0.15. \quad (130)$$

Each fresh panel contains 256 attempted cases. Divide the depth interval logarithmically into four parts, divide the allowed total amplitude $\sum_j a_j/h$ into a lower and an upper half, and distinguish unidirectional cases, in which all s_j agree, from bidirectional cases, in which both signs occur. Attempt four cases in every combination of the four depth intervals, four mode counts, two amplitude halves, and two direction classes. A deterministic seeded PCG64 stream selects the remaining continuous quantities within each category. The stream identifier and all modes, amplitudes, phases, directions, case indices, and realized parameters are stored.

The development and final panels use the respective seeds 2026072201 and 2026072202. Rejection never causes replacement. At least two of the four attempts in every category must pass the common reference checks; otherwise the panel construction fails. The final panel remains untouched until the corpus, model, optimizer, and checkpoint rule are frozen.

The 11 historical shallow initial conditions are also retained as a separate development panel, but their trajectories are recomputed with the same frozen reference solver used everywhere else; the old trajectory files are historical evidence only. All sparse-Fourier rollouts have final time 20 and are stored every 0.08 time units. An earlier unmatched comparison suggested that shallow packets reduced the shallow error tail, but simultaneous changes in corpus size, normalization, and device count prevent a causal claim. That observation motivates this held-out test, not a sparse-Fourier training family.

C Rescaling to the 2π reference torus

The period dependence of G is suppressed in (8). In this appendix only, write $G^{[L]}(\eta; h)$ for the physical DNO on $\mathbb{T}_L$. Set $\lambda = L/(2\pi)$ and write $x = \lambda X$, $y = \lambda Y$. Define

$$\bar{\eta}(X) = \frac{\eta(\lambda X)}{\lambda}, \quad \bar{\xi}(X) = \xi(\lambda X), \quad \bar{h} = \frac{h}{\lambda}. \quad (131)$$

$$\boxed{[G^{[L]}(\eta; h)\xi](\lambda X) = \frac{1}{\lambda}[G^{[2\pi]}(\bar{\eta}; \bar{h})\bar{\xi}](X).} \quad (132)$$

Indeed, $\bar{\varphi}(X, Y) = \varphi(\lambda X, \lambda Y)$ solves the same Laplace–Neumann–Dirichlet problem below $\bar{\eta}$ at depth $\bar{h}$. The chain rule gives $\bar{\varphi}_X = \lambda \varphi_x$, $\bar{\varphi}_Y = \lambda \varphi_y$, and $\bar{\eta}_X(X) = \eta_x(\lambda X)$, which proves (132) directly.

The cutoff in this document is a physical wavenumber cutoff. Let $P_K^{[L]}$ denote projection on $\mathbb{T}_L$ onto integer Fourier indices $j \in \mathbb{Z}$ satisfying $|2\pi j/L| \leq K$, and let $P_{\lambda K}^{[2\pi]}$ denote the corresponding projection on $\mathbb{T}_{2\pi}$. For $\bar{f}(X) = f(\lambda X)$,

$$(P_K^{[L]} f)(\lambda X) = P_{\lambda K}^{[2\pi]} \bar{f}(X). \quad (133)$$

Write $G_{\text{ref}, K}^{[L]}$ for the operator in (11) evaluated on $\mathbb{T}_L$ with cutoff K , and write $G_{\text{ref}, \lambda K}^{[2\pi]}$ for the corresponding operator on $\mathbb{T}_{2\pi}$. If the same Fourier indices are present on both N -point grids, then, up to floating-point roundoff,

$$\lambda[G_{\text{ref}, K}^{[L]}(\eta; h)\xi](\lambda X) = [G_{\text{ref}, \lambda K}^{[2\pi]}(\bar{\eta}; \bar{h})\bar{\xi}](X). \quad (134)$$

For Fourier index j , $k_j = 2\pi j/L = j/\lambda$, whereas the corresponding reference-torus wavenumber is $\bar{k}_j = j$. Define $\bar{a} = a/\lambda$, $\bar{H}_s = H_s/\lambda$, and $\bar{k}_p = \lambda k_p$. Then $k_j h = \bar{k}_j \bar{h}$, $k_j a = \bar{k}_j \bar{a}$, $a/h = \bar{a}/\bar{h}$, $H_s/(2h) = \bar{H}_s/(2\bar{h})$, and $k_p H_s/2 = \bar{k}_p \bar{H}_s/2$. Thus each dimensionless combination used in the family definitions is unchanged by a uniform rescaling of length.

The production corpus remains on $L = 2\pi$. Equations (132)–(134) are pre-normalization identities for DNO evaluation; they do not by themselves define a variable-period training augmentation. A full trajectory rescaling at fixed gravity uses $\tau = t/\sqrt{\lambda}$. The rescaled fields are

$$\begin{aligned} \eta^\sharp(X, \tau) &= \lambda^{-1} \eta(\lambda X, \sqrt{\lambda} \tau), & \xi^\sharp(X, \tau) &= \lambda^{-3/2} \xi(\lambda X, \sqrt{\lambda} \tau), \\ [G^{[2\pi]}(\eta^\sharp(\cdot, \tau); \bar{h})\xi^\sharp(\cdot, \tau)](X) &= \lambda^{-1/2} [G^{[L]}(\eta(\cdot, \sqrt{\lambda} \tau); h)\xi(\cdot, \sqrt{\lambda} \tau)](\lambda X). \end{aligned}$$

It must therefore be treated as a separate change of variables rather than an archive-level spatial resize.

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